What Makes a Latent Fire? Discovering Causal Dependencies Between Sparse Autoencoder Latents

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Abstract

Circuit discovery typically explains an output: a logit, a probability, a behavioural metric. We ask instead what makes an internal latent fire. For a target sparse autoencoder (SAE) latent, *weighted circuit masking* (WCM) learns a circuit of upstream latents with a scaling coefficient for each, trained so that the circuit alone reproduces the target under three different ablations of everything else. At a sparsity penalty that yields circuits of 10^2–10^3 upstream latents, 46% of ∼15,000 sampled latents in TuringLLM, a 12-layer language model, get a circuit that reproduces the target under all three ablations and is necessary for it, on held-out text [*pending: Pythia-70m and GPT-2 small, DAN-40/41*]. A weaker penalty gives larger circuits that pass more often. The coefficients are what keep these circuits small: without them, circuits for targets in the last four layers need $\sim 10^4$ latents to pass, about 10× more, and only half as many of those targets pass at all. Without their coefficients, the same nodes pass for 1.5% of targets instead of 45%. Random circuits of the same size, given the same coefficient fitting, failed on all 747 targets tested. The procedure needs no behavioural task and applies to any latent in an SAE bank.

1 Introduction

Consider a latent in layer 10 of a 12-layer language model that fires on the possessive in "Ohm's", "Snell's", or "Kirchhoff's", before the word "law" has appeared, but stays silent on "Darwin's", "Rutherford's", and "Picasso's". Given an invented name, it fires in a passage on electromagnetism and not in one on the history of painting. Something upstream supplies what the model knows about which names carry laws, and what the passage is about. What makes this latent fire?

Sparse autoencoders (SAEs) decompose a model's activations into latents like this one, which often track coherent concepts (Cunningham et al., 2024; Bricken et al., 2023; Gao et al., 2024; Templeton et al., 2024), a working answer to superposition (Elhage et al., 2022). A latent says what the model represents; it does not say how the model computes it. That question belongs to circuit discovery, whose methods, over components or features and by search, attribution, or learned masks, predominantly target an output (Conmy et al., 2023; Syed et al., 2023; Marks et al., 2025; Ameisen et al., 2025; Bhaskar et al., 2024; Caples et al., 2025). An output is the model's computation expressed in our vocabulary; targeting latents instead can capture variables that never show up as a token.

Latent targets have been approached by attribution, tracing upstream of a chosen SAE latent and thresholding the scores (Ge et al., 2024; Golimblevskaia et al., 2025). To our knowledge, no method *optimises* a circuit for a single target latent and tests it by intervening on that latent. We call this *latent-targeted circuit discovery* (Figure 1).

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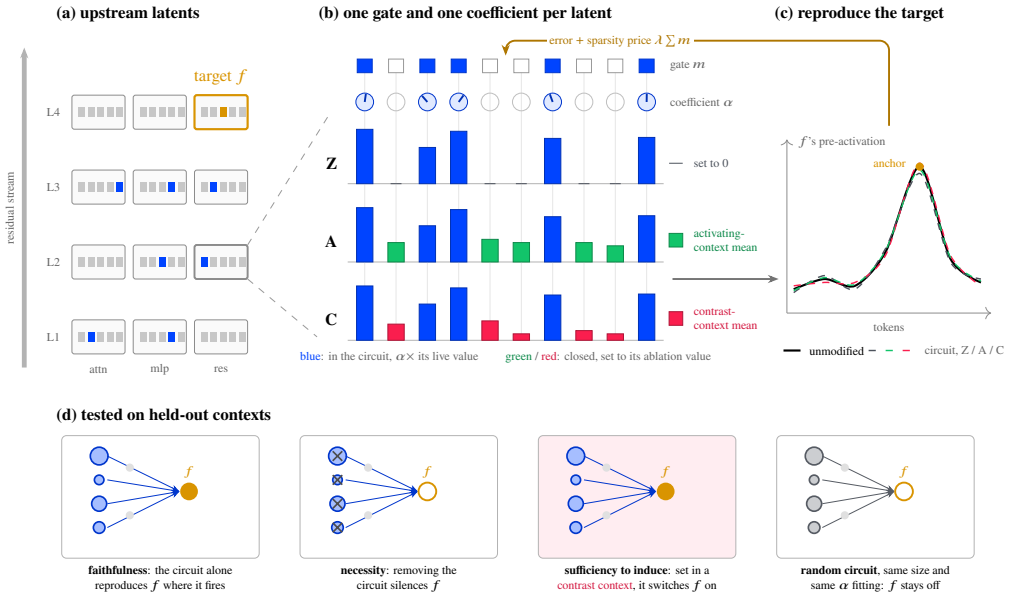


Figure 1: How weighted circuit masking works. **(a)** For a target latent f, every latent at every upstream SAE site is a candidate node. **(b)** Each candidate gets a gate m (kept or closed) and a scaling coefficient α: a kept latent contributes α times its live value, and a closed one is replaced by its ablation value. **(c)** They are trained so that, under each ablation method, the circuit reproduces f's pre-activation at its anchor positions, with a sparsity penalty on every open gate. **(d)** The weighted circuit, typically 10^2–10^3 latents, is tested on held-out contexts by faithfulness, necessity, and sufficiency to induce, against same-size random circuits with fitted coefficients.

Our method, *weighted circuit masking* (WCM), learns two things for every upstream latent: a gate for whether it belongs to the circuit, and a scaling coefficient for how strongly it acts. Both are trained so that the circuit alone reproduces the target with everything else ablated in three ways at once: set to zero, or to its mean over contexts where the target fires, or where it is silent. We test each circuit on held-out text for *faithfulness* (the circuit alone reproduces the target), *necessity* (removing the circuit removes the target), and *sufficiency to induce* (setting the circuit's values switches the target on where it was silent).

We apply WCM to ~15,000 sampled latents of TuringLLM, a 12-layer model trained on educational text [*pending: Pythia-70m and GPT-2 small, DAN-40/41*]. We fit every target at one sparsity penalty, which yields circuits of 10^2–10^3 upstream latents. At that penalty, 46% of targets get a circuit that passes a strict joint test: faithful under all three ablations and necessary, on held-out text. The rate is one point on a size trade-off, not a ceiling: in a 190-target sweep it rises from 42% to 58% as a weaker penalty grows the median circuit from about 500 to 1,100 latents (Appendix E) [*pending: weaker penalties, DAN-131*]. Residual-stream targets pass most often (65%) and attention outputs least, about one in seven. The coefficients are what keep circuits small: without them, circuits for targets in the last four layers need ~10^4 latents to pass, 10× more, and only 41% of those targets pass at any size we tried, against 80% with coefficients (Section 5.2). The node sets themselves are not unique, since equally faithful refits share only part of their nodes, so we treat each as one member of an equivalence class (Section 5.6); and the study covers Top-K dictionaries only (Section 6.1). Section 5.3 opens one passing circuit, for a latent that reads a date or place as classical Greek antiquity, and Appendix I returns to the law latent above.

We contribute:

- **Weighted circuit masking** (Section 4.3): a gate and a separate scaling coefficient for each upstream latent, trained under three ablations at once; the coefficients shrink passing circuits by

about 10× in deep layers and double the share of deep targets that pass.
- **Latent circuits at scale** (Section 5.1): circuits for ∼15,000 sampled latents, tested on held-out text; passing circuits hold 10^2–10^3 latents, and success varies systematically with site and depth.
- **An evaluation protocol for latent targets** (Sections 4.4 and 5.5): each score names its ablation, random baselines receive the same coefficient fitting, and each method wins only the scores it was trained on, so we train on all three.

We release the pipeline, discovery methods, and analysis code at `github.com/DanielJamesDavies/turing-explorer-circuit-discovery`.

2 Related Work

Circuit discovery. Manual analysis of model components (Olah et al., 2020; Wang et al., 2023) gave way to automated search (Conmy et al., 2023) and to attribution patching and its integrated-gradient and relevance-based refinements (Nanda, 2023; Syed et al., 2023; Kramár et al., 2024; Hanna et al., 2024; Rezaei Jafari et al., 2025). Sparse feature circuits (Marks et al., 2025) moved the nodes to SAE features with error nodes for the reconstruction residual, and attribution graphs (Ameisen et al., 2025; Lindsey et al., 2025) build per-prompt graphs over transcoder features at production scale. Their objectives are anchored at an output, and circuits for one task can differ across its prompts (Makou et al., 2026).

Circuits for internal features. Internal targets are not new. Hierarchical attribution extracts the subgraph behind a chosen SAE feature (Ge et al., 2024), dictionary features have been traced patch-free in Othello-GPT (He et al., 2024), transcoder features are reverse-engineered from their input circuits (Dunefsky et al., 2024), latent attribution debugs a chosen latent (Dupre la Tour and Mossing, 2025), CircuitLens explains a feature's activation from upstream interactions (Golimblevskaia et al., 2025), and attribution graphs have been extended to concept directions read by linear probes (Palit et al., 2026). Upstream features have also been ranked by single ablations (Cunningham et al., 2024) or linked by sparse regression (Yin et al., 2026). These methods attribute, ablate one node at a time, or regress; circuit probing is closer in spirit, optimising a mask over model weights for a labelled intermediate variable (Lepori et al., 2023), and sparsely connected transcoders learn latent-to-latent masks while the dictionary is trained (jacob_drori, 2025).

Masks and scalars. Learned masks descend from subnetwork probing (Cao et al., 2021; Conmy et al., 2023) and edge pruning (Bhaskar et al., 2024; Li and Janson, 2024); over SAE latents they have localised attributes (Chaudhary and Geiger, 2024) and found behavioural circuits (Caples et al., 2025), the latter being our unweighted circuit masking with its objective moved to a latent. Such masks keep a node at its natural scale or ablate it; learned per-unit scales appear with other jobs, from pruning criteria and post-pruning rescaling (Liu et al., 2017; Kwon et al., 2022) to steering and regression weights (Stoehr et al., 2024; Yin et al., 2026), and gate-times-magnitude parametrisations separate membership from strength outside circuit discovery (Louizos et al., 2018; Lai et al., 2025) (Appendix J).

Evaluating circuits. What removing a latent means is an experimental choice that can decide the result (Li and Janson, 2024; Miller et al., 2024; Mueller, 2024); patching in clean values (denoising) and removing them (noising) answer different questions (Heimersheim and Nanda, 2024), and our sufficiency to induce is a denoising-style test. Prior work evaluates under several ablations but discovers under one; the nearest objectives sum two directions of one counterfactual (Chen et al., 2025) or certify bounded patching perturbations (Hadad et al., 2026). Random circuits as a null for faithfulness follow the hypothesis tests of Shi et al. (2024), and MIB standardises circuit evaluation for output targets (Mueller et al., 2025).

What this paper adds is the conjunction: circuits of SAE latents *optimised* for a single target latent, with a gate and a separate *fitted* coefficient per node, trained under three ablations at once, and

tested on held-out contexts against random circuits given the same fitting, for ∼15k sampled latents per model [*pending: only TuringLLM so far; Pythia-70m and GPT-2 small, DAN-40/41*] (further related work in Appendix J).

3 Setting

The primary model is TuringLLM-1.0-254M, a 12-layer decoder-only transformer trained from scratch on a documented synthetic corpus of ∼2 billion tokens generated by Phi-3 Mini (Abdin et al., 2024); a model whose training data is known is a more tractable subject than one trained on an opaque crawl (Appendix A). A bank of 36 Top-K sparse autoencoders, one per site (attention output, MLP output, and residual stream at each layer), maps each activation onto 40,960 latents with $K = 128$ active per position (Appendix A.4). The same protocol is applied to Pythia-70m and GPT-2 small with public Top-K dictionaries (Section 5.1) [*pending: public-model runs, DAN-40/41*]. As a running example we follow a layer-9 residual-stream latent of TuringLLM that fires where a passage places its subject in classical Greek antiquity: after "In the 5th century BC, the philosopher taught in the city of" it reaches 76% of its typical activation, and after "In the 19th century, . . . " it stays silent (Section 5.3).

4 Method

For a latent f and a sequence–position pair (x, t), write $a_f(x, t) \geq 0$ for its activation (after Top-K) and $\hat{a}_f(x, t)$ for its pre-activation (before it); pseudocode is in Appendix L.

4.1 Problem

Targets and contexts. Fix a target latent f and let U_f be the latents at every SAE site that causally precedes f's site. Each target has *activating contexts* from two stores, up to 64 of its strongest-activating sequences and up to 64 drawn from the middle of its activation distribution, each with an anchor at the target's peak (Appendix M). With full stores it trains on 32 strongest and 16 mid-band contexts and is scored on 16 held-out strongest contexts (primary) and 16 held-out mid-band contexts (reported); the split is stratified by activation and always trains on the two strongest (Appendix B). It also has 64 *contrast contexts*: corpus sequences ranked by their highest cosine similarity to any of its strongest-activating contexts in pooled final-layer representation space, excluding its activating contexts, and kept only if the target stays out of its site's Top-K on them, so that they are close to the target's contexts but verified silent (Appendix M). For the running example they are history passages about other periods, such as the Enlightenment, Renaissance music, or the pre-Columbian era. Targets are drawn uniformly at random, the same number per site, with no prioritisation heuristic (Appendix B).

Circuit-only execution. A *weighted circuit* is a node set $C \subseteq U_f$ with a scaling coefficient $\alpha_{s,i} > 0$ for each node. Under an ablation method π, the forward pass is modified at every upstream site s: the site's Top-K code c_s, computed from the (already modified) stream, is replaced by

$$\hat{c}_{s,i} = \begin{cases} \alpha_{s,i}\, c_{s,i} & (s,i) \in C \\ v^{\pi}_{s,i} & \text{otherwise,} \end{cases}$$

and the stream continues as $x + W^s_{\text{dec}}(\hat{c}_s - c_s)$, so SAE errors pass through unchanged and every downstream site, including the target's, reads the modified stream. Nodes are *recomputed*, not clamped: a node's value at each position is whatever the modified stream gives it. The ablation value v^{π} is zero (Z), the latent's mean over activating contexts (A), or its mean over contrast contexts (C), optionally sparsity-preserving, with the per-position Top-K re-imposed. An unweighted circuit fixes $\alpha = 1$. Writing a_C for the target's mean activation at the anchors under this intervention, $a_\varnothing$ for the

empty circuit, and a_{pos} for the unmodified model, *faithfulness* is

$$\phi_\pi^{\text{free}}(C) = \frac{a_C - a_\varnothing}{a_{\text{pos}} - a_\varnothing},$$

and *clamped faithfulness* $\phi_\pi^{\text{pin}}(C)$ is the same ratio with nodes clamped to their clean-run values (times α).

The problem. Faithfulness depends on the ablation method (Miller et al., 2024; Mueller, 2024), and a node set trained under one method can be exactly infeasible under another (Section 4.3), so one weighted circuit is asked to serve all three:

$$n_\varepsilon(f) \;=\; \min\big\{\, |C| \;:\; \exists\, \alpha \;\text{ with }\; \phi_\pi^{\text{free}}(C, \alpha) \geq 1 - \varepsilon \;\text{ for every } \pi \in \{\text{Z}, \text{C}, \text{A}\} \,\big\}. \tag{P}$$

WCM solves a differentiable relaxation of (P). With $\alpha = 1$ the smallest faithful node sets are several times larger, and attribution at natural scale needs one to two orders of magnitude more (Section 5.2); (P) asks for a minimiser: its near-optimal solutions form a large, non-nested equivalence class (Section 5.6), so only class-invariant quantities are treated as properties of the target.

4.2 Attribution Baselines

WCM is compared against rank-and-cut attribution. Every upstream latent is scored by a first-order estimate of its effect on the target: a *contrastive* objective on contrast contexts, taken through the target's pre-activation because Top-K censors its activation to zero there; an *ablation* objective on activating contexts, where the activation is live; and their fusion. Each runs under four linearisation modes: *local* is attribution patching with a zero patch and the gradient taken on the clean run, the AtP form of Kramár et al. (2024), and *path attribution* is an integrated-gradients estimator in the style of sparse feature circuits (Marks et al., 2025), taken jointly over sites from their mean-ablation values (differences in Appendix C). Scores are cut per position and the node set is the union over the target's causal prefix, since a Top-K code is sparse per position while the target reads every position; the 128-target grid of Appendix Q.1 predates this construction and uses per-site top-K caps instead (Appendix L). Objectives, modes, and the node-set construction are in Appendix C.

4.3 Weighted Circuit Masking

Rather than thresholding attribution scores, WCM learns the circuit directly, adopting the optimisation framing of learned-mask circuit discovery (Bhaskar et al., 2024; Caples et al., 2025) with a continuous-sparsification gate (Savarese et al., 2020), and moving its objective from an output to a single latent. For every upstream site s it attaches gate parameters $\theta_s \in \mathbb{R}^{d_{\text{sae}}}$ with gates $m_s = \sigma(\theta_s)$, and coefficient parameters ψ_s with $\alpha_s = \text{softplus}(\psi_s)$. A latent is kept at α times its live value when its gate is open and sent to its ablation value when closed:

$$x \;\leftarrow\; x + W_{\text{dec}}^s \big[\, m_s \odot \alpha_s \odot c^s + (1 - m_s) \odot c_{\text{abl}}^s - c^s \,\big],$$

where c^s is the site's dense Top-K code and c_{abl}^s its ablation value, a dense mean in training (the sparsity-preserving variant is used only in evaluation); editing only the decoded difference leaves the SAE error untouched, as in evaluation.

Objective. The circuit is trained to *reproduce* the target's pre-activation at the anchors of the training contexts, by mean squared error against its value on the unmodified stream. The pre-activation is forced by Top-K: the activation is censored at zero, so a loss on it carries no gradient exactly where the mask must learn. Reproducing the pre-activation does not, however, keep the target in its site's Top-K: the loss constrains the site's stream along one direction of d_{model}, and a circuit can restore that projection while other latents at the site rise past the target. A *rank-keep* term therefore asks the

target to hold its natural rank: with r the target's rank among its site's other latents at an anchor in the unmodified model, capped at K, and $\tau_\pi^{(r)}$ the r-th largest of their pre-activations under the circuit,

$$\mathcal{R}_\pi = \text{mean}\Big[\text{relu}\big(\tau_\pi^{(r)} - \hat{a}_f\big)^2\Big],$$

which is zero whenever the target is at or above its natural rank, so it never pulls a circuit towards a state the unmodified model does not have.

Training under three ablation methods. A mask trained under one ablation method learns whatever that method rewards. Trained only under mean ablation over contrast contexts, it reaches high faithfulness under that ablation while its zero-ablation faithfulness is 0.0 at depth: it learns the difference from the contrast-context mean, never a set that suffices alone. Trained only under zero ablation, it suffices alone but is never asked what separates firing from a near-identical context that does not fire. WCM therefore scores one mask under all three ablation methods at once, with gradients accumulating into the same parameters:

$$\mathcal{L} = \frac{\sum_\pi \gamma_\pi \big(\mathcal{L}_\pi + \gamma_\text{R}\, \mathcal{R}_\pi\big)}{\nu} \;+\; \lambda \sum_s \sum_i \sigma(\theta_{s,i}), \qquad \nu = \text{mean}\big(\text{target}^2\big),$$

summed over $\pi \in \{\text{Z}, \text{C}, \text{A}\}$, with $\gamma_\text{Z} = 1$, $\gamma_\text{C} = \gamma_\text{A} = 0.25$ at every layer, and $\gamma_\text{R} = 3 \times 10^{-3}$; the ablation means are computed on training-split contexts only (Appendix B). All terms measure the same quantity on the same target, so one shared, bounded normaliser suffices, and the sparsity penalty is flat per latent, which outperformed every gradient-informed site weighting we tried (Appendix P). In calibration the A term roughly halves the smallest passing node set on layer-2 targets and raises the share of layer-9 targets with any passing circuit. The ablation method matters more than one might expect: on four layer-2 targets, node sets trained under different ablation methods at matched size share fewer latents (Jaccard 0.13) than node sets trained for different *objectives* under the same method (0.54).

Scaling coefficients. Coefficients start at $\alpha = 1$, so training starts exactly at unweighted behaviour; *unweighted circuit masking* keeps them there. One guard is required: with coefficients free, the optimiser can hold a gate just below the keep threshold while inflating its coefficient, emptying the node set while keeping function intact, so latents with closed gates are charged $\lambda\,(1 - \sigma(\theta))\,|\alpha - 1|$. The zero-ablation term anchors the coefficients: trained without it, they calibrate against the mean ablation's background and overshoot by one to three orders of magnitude once that background is removed (Appendix E).

Binarisation. Evaluation scores the hard node set $\{i : m_i > \frac{1}{2}\}$, so the gates are annealed during training, $m = \sigma(\theta/T)$ with T decaying geometrically from 1 to 0.05 (continuous sparsification; Savarese et al., 2020), and the training forward converges to the binary gates that evaluation scores. The gates are optimised directly rather than sampled (Louizos et al., 2018; Li and Janson, 2024); the local minima that sampling avoids appear here as the equivalence class of near-optimal node sets (Section 5.6). Training runs 400 AdamW steps; all settings are in Appendix B, and the binarisation and size-calibration studies in Appendix P.

4.4 Evaluation

Table 1 names the evaluation family; every reported number carries its row, is read on the held-out activating contexts (or held-out contrast contexts), and reads the target's activation in every term of its ratio (Appendix B).

Score	Contexts	Circuit latents	Other latents
ϕ_{ind}	contrast	set	untouched
ϕ_{sup}	activating	zero	recomputed
ϕ^{pin}	activating	clamped to $\alpha\times$ clean values	zero / activating means
$\phi^{\text{free}}_{\text{Z}}$	activating	recomputed (permitted)	zero
$\phi^{\text{free}}_{\text{A}}$	activating	recomputed	activating means, dense
$\phi^{\text{free}}_{\text{A,sp}}$	activating	recomputed	activating means, sp
$\phi^{\text{free}}_{\text{C,sp}}$	activating	recomputed	contrast means, sp
$\phi^{\text{free}}_{\text{Z},\alpha}$ / $\phi^{\text{free}}_{\text{A},\alpha}$	activating	recomputed, $\times\,\alpha$	zero / activating means
$\phi^{\alpha}_{\text{ind}}$	contrast	set to $\alpha\times$ anchor means	untouched

Table 1: The evaluation family. Intervention scores: ϕ_{ind} sufficiency to induce, ϕ_{sup} necessity; "set" means set, at every position, to the node's mean value at the anchors of the activating contexts. Faithfulness ϕ^{free}_{π} measures whether the circuit reproduces the target alone with its nodes recomputed, clamped faithfulness ϕ^{pin} whether it contains the right latents, and each variant names its ablation method π (Z zero, A mean over activating contexts, C mean over contrast contexts; sp sparsity-preserving; Section 4.1). The α rows score weighted circuits with their scaling coefficients applied (Section 5.2). SAE errors are preserved in every row, and every ratio reads the target the same way in all of its terms (Appendix B).

Intervention scores. Let A^+ and A^- be the target's mean activation on activating and contrast contexts. **Sufficiency to induce**, $\phi_{\text{ind}} = (A^-_{\text{int}} - A^-)/(A^+ - A^-)$, sets the circuit's nodes in contrast contexts to their mean values at the anchors of the activating contexts, times α for a weighted circuit ($\phi^{\alpha}_{\text{ind}}$, the form reported throughout): near 1, the circuit switches the target on. **Necessity** is one minus the completeness score of Marks et al. (2025): with $a_{M\setminus C}$ the target's activation on activating contexts when the circuit's nodes are set to zero and every other latent is recomputed, and $a_{\varnothing}$ the zero-ablation empty circuit,

$$\phi_{\text{sup}}(C) = \frac{a_{\text{pos}} - a_{M\setminus C}}{a_{\text{pos}} - a_{\varnothing}}.$$

Two details differ from their definition: SAE errors are always preserved here, where they ablate the circuit's error nodes, and the nodes are zero-ablated rather than mean-ablated, so the score needs no reference distribution. Both scores intervene on the circuit as one set. All scores are unclipped ratios: success means landing near 1, and a score far above 1 is overshoot, not quality.

Ablation methods. The choice of ablation is load-bearing (Li and Janson, 2024; Miller et al., 2024). Zero ablation needs no reference distribution. Mean ablation over activating contexts leaks target signal into the ablation value increasingly with depth ($a_{\varnothing}$ up to 30% of a_{pos}); mean ablation over contrast contexts does not. Dense mean ablation puts the model on streams a Top-K SAE can never emit, so the sparsity-preserving variant re-imposes the per-position Top-K after the means are applied (Appendix N.2).

Random-circuit baseline. Fitted coefficients could make any set look faithful, so each random circuit, of the same size and drawn from latents active on the target's strongest training contexts at the same sites, receives the *same coefficient-fitting budget* on its own nodes before scoring; faithfulness then reflects selection rather than the fitting capacity of the coefficients.

Specificity. Every score above reads the target alone, so a circuit that restores the target by lifting its whole neighbourhood at the site passes them all. We therefore read the full code at the target's site under the circuit and report, as a measured property rather than a pass criterion: the target's rank and Top-K membership, the number of latents the circuit lifts above the unmodified model's Top-K cut relative to the empty circuit, and the faithfulness of *sibling* latents that co-fire with the target, against the target's own (Appendix H).

Near-threshold targets. A target that only just enters its site's Top-K in the unmodified model has an ill-conditioned activation read: a circuit can reproduce its pre-activation while small shifts near the cut push it out, so its activation-read scores are near zero whatever the circuit. Such targets, those whose median rank among all their site's latents at the held-out anchors is at least $K/2$, are identified from the unmodified model alone, before any circuit is fitted; they stay in the headline pass rate and are also reported as a separate stratum on both reads.

Pass criterion. A circuit passes if, on the held-out strongest contexts and the activation read, its faithfulness under all three ablation methods (Z, sparsity-preserving A, and C) lies in $[0.8, 1.5]$ and its necessity is at least 0.9. The band is asymmetric by design: below 0.8 the circuit fails to drive its target, while an overshoot up to 1.5 still drives it, and whether that overshoot is specific to the target is measured for every circuit (above). Each circuit's largest deviation $\max_\pi |\phi_\pi^{\text{free}} - 1|$ is reported alongside the binary rule; sufficiency to induce, specificity, and the random-circuit baseline are reported rather than required; targets whose zero-ablation denominator is vacuous ($|a_{\text{pos}} - a_\varnothing| < 0.05\, a_{\text{pos}}$) are excluded and counted, and an undefined A or C ratio counts as a fail.

5 Experiments and Results

Setup. Results come from three sources. The *production run* fits WCM[Z+C+A] to 15,046 sampled TuringLLM targets under protocol v1 (Appendix B); [*pending: Pythia-70m and GPT-2 small, DAN-40/41*]. A depth-stratified *panel* of 22 TuringLLM targets carries the matched-size comparisons against other discovery methods; its current numbers predate protocol v1 and are marked as such. A 128-target *grid* of an earlier form of the gradient attribution methods grounds the contrast-context and co-activation findings (Appendix Q.1).

5.1 Weighted Circuits at Scale

Figure 2 breaks the production run down by target layer and site: 15,004 fitted circuits (42 targets had too few activating or silence-checked contrast contexts, and 10 circuits with a vacuous denominator are excluded). Under the pass criterion (Section 4.4), 46% of circuits pass at the production sparsity penalty. They hold a median of 494 upstream latents (386 for passing circuits), and 91% lie within 10^2–10^3 (Figure 16). The pass rate depends on the target's site and depth: 65% of residual-stream, 52% of MLP, and 14% of attention-output targets pass, and the rate falls from 85% at layer 1 to 15% at layer 11. The figure shows where the fall comes from. Zero ablation and necessity both hold for about four circuits in five at every depth (79%, 80%, and 76% at layers 0–4, 5–7, and 8–11); what grows with depth is the share that fail only under the two mean ablations, from 10% of circuits at layers 0–4 to 53% at 8–11 (median faithfulness 0.64 and 0.48 at layer 11), and these make up 59% of all failing circuits. Attention outputs are the exception: 40% of their circuits fail even zero ablation, and 63% of those belong to near-threshold targets, which barely enter their site's Top-K in the unmodified model and read 0 once they slip out of it (1,142 targets, 7.6% of the run, almost all attention outputs; 0.8% pass, against 50% for the remaining targets). Layer 0 departs from the trend (55%): its circuits are faithful (90%), but a third fail necessity, because the token embedding reaches layer-0 targets directly and is never ablated. On the held-out mid-band contexts, where targets fire more weakly, 12% pass (median largest deviation 0.77, against 0.31 on the strongest contexts), so circuits fitted mostly to strong contexts transfer only in part to weak ones. Specificity is measured rather than required: 30% of circuits (44% at MLP targets) are flagged as amplifiers, restoring co-firing sibling latents to at least 80% of the target's faithfulness under some ablation method (Appendix H). Sufficiency to induce with coefficients, $\phi_{\text{ind}}^{\alpha}$, which no circuit is trained on, is reported separately (Figure 17): its median is 0.98 among passing circuits, and over all circuits 1.19 at residual-stream and 0.20 at attention-output targets. Sizes are at a fixed 400-step budget, at which most circuits are still shrinking (Figure 18).

Beyond TuringLLM. The same protocol is applied unchanged to two public models with public *Top-K* dictionaries: EleutherAI's k=16, 32,768-latent dictionaries for Pythia-70m and OpenAI's

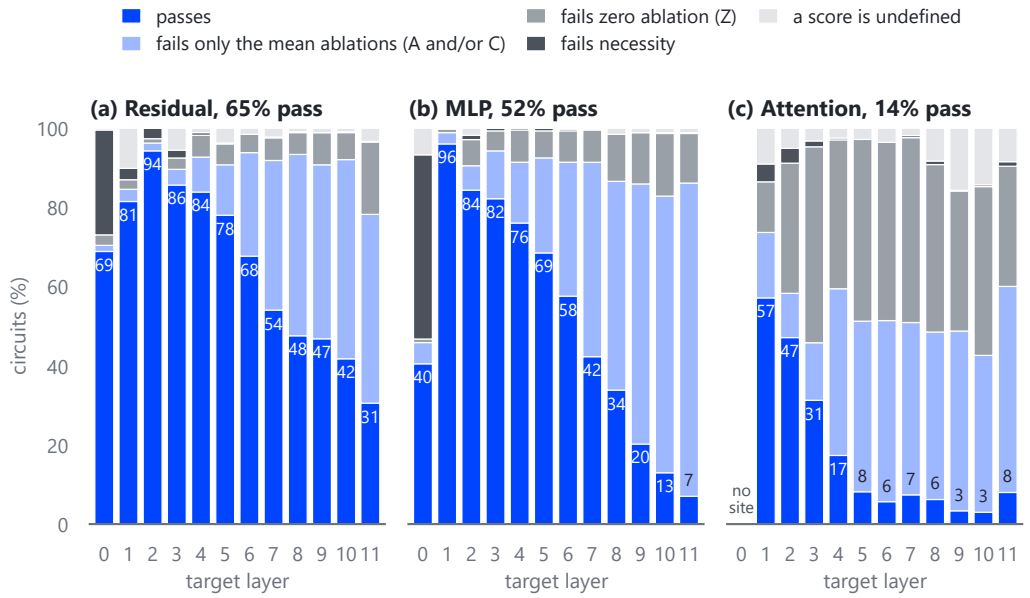


Figure 2: How the weighted circuits of 15,004 TuringLLM latents fare at the production sparsity penalty, by target layer and site kind. Each circuit is counted once: it passes the pre-stated criterion (faithfulness under Z, A, and C in $[0.8, 1.5]$ and necessity ≥ 0.9, on held-out contexts); fails only the mean ablations (A, C, or both) while zero ablation and necessity hold; fails zero ablation while necessity holds; fails necessity; or has an undefined score (3%, counted as a fail). Numbers on the bars are pass rates (%), and panel titles give each site kind's overall pass rate; layer-0 attention outputs have no upstream site.

k=32, 32,768-latent dictionaries for GPT-2 small, at every layer (sites in Table 4). Only the model and dictionary adapters differ; discovery, the ablation methods, the contrast contexts, and every score run through the same code as on TuringLLM. [*pending: full-run results, DAN-40 and DAN-41, folded into the figure above*]

5.2 Why the Coefficients Matter

Three versions of each circuit separate what the nodes contribute from what the coefficients contribute: the weighted circuit as fitted, the same nodes at natural scale ($\alpha = 1$), and a random circuit of the same size given the same coefficient fitting (Section 4.4). On 1,746 circuits sampled from the production run, the weighted circuits pass for 44.7% of targets and the same nodes at natural scale for 1.5%: median faithfulness falls from 0.94, 0.95, and 0.89 to 0.00, 0.09, and 0.06 under Z, A, and C, and at natural scale no depth band passes more than 4% of its targets (Figure 11). Random circuits given the same coefficient fitting pass for none (below). At natural scale the members carry the target only part of the way: with every coefficient set to 1 they restore a median 0–19% of its activation, against 90–97% with the fitted coefficients, and the coefficients do this without returning the members to their own natural values (Figure 12). [*pending: the same comparison on Pythia-70m and GPT-2 small, DAN-40/41*] At natural scale, attribution needs 10^4–10^5 latents before the circuit alone reproduces a deep target (Appendix D); the coefficients locate most of that cost in redundancy, many latents each carrying a small share of a signal that fewer, rescaled nodes carry alone. Unweighted circuit masking isolates what the coefficients buy: the same objective and optimiser with α held at 1. Figure 3 follows 190 targets across a sweep of sparsity penalties and asks, for a budget of N latents, what share of targets have a passing circuit that small. Weighted circuits explain more targets at every budget, and

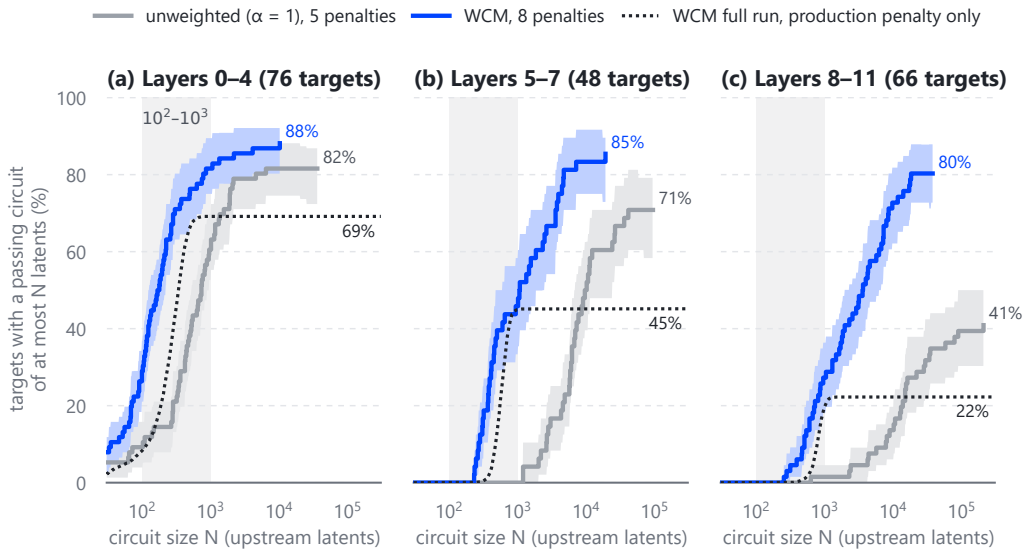


Figure 3: Fitted coefficients explain more targets at every circuit budget. For each target depth band, the share of 190 sweep targets with a circuit of at most N upstream latents that passes the criterion of Section 4.4 (faithfulness under Z, A, and C in $[0.8, 1.5]$ and necessity ≥ 0.9, on held-out contexts), taking each target's smallest passing circuit over the sparsity penalties it was fitted at: weighted circuit masking (blue) at eight penalties, λ from 4×10^{-3} to 2.5×10^{-5}, and unweighted circuit masking ($\alpha = 1$, grey) at five, 10^{-3} to 10^{-5}. Targets that never pass count as unexplained; shading is a 90% bootstrap interval over targets, and each curve ends at the largest circuit fitted. Dotted: the full run (~15,000 targets) at the production penalty alone, which levels off at that penalty's pass rate.

keep explaining more as the budget grows: within 10^3 latents they pass for 82%, 48%, and 27% of targets at layers 0–4, 5–7, and 8–11, against 63%, 0%, and 2% for unweighted circuits, and over the whole sweep for 88%, 85%, and 80%, against 82%, 71%, and 41% even with unweighted circuits of up to $\sim 2 \times 10^5$ latents. Where both methods pass, the smallest unweighted circuit is a median 3.7× larger at layers 0–4, 9.6× at 5–7, and 10.5× at 8–11, where it holds $\sim 10^4$ latents (13,075 against 906). The production penalty is one point on the weighted curves: the full run, fitted at that penalty alone, levels off at 69%, 45%, and 22% (dotted), while weaker penalties carry the same targets much further. At that shared penalty unweighted circuits keep 1.85× as many latents and pass 54% against 64% at layers 0–4, but 2.1% against 38% at 5–7 and 1.5% against 18% at 8–11 (Figure 10): their scores are lower one by one, and the criterion asks for all of them at once (Appendix E).

For the running example, the weighted circuit passes with 728 latents, while unweighted circuit masking on the same target fails at every penalty we tried, up to 24,837 latents, where its activating-mean faithfulness is still 0.71.

Random circuits. Random circuits of the same size as each target's circuit, drawn from latents active on its strongest training contexts and given the same coefficient fitting, failed on all 747 targets of a stratified sample, against 43% for the fitted circuits on the same targets. Their median faithfulness is 0.00 under every ablation method at every depth, only one has even a single score in the pass band, and none induces the target to half its level in contrast contexts, so faithfulness and sufficiency to induce are properties of the selected node set, not of the fitting capacity of the coefficients. [*pending: the same baseline on Pythia-70m and GPT-2 small, DAN-40/41*]

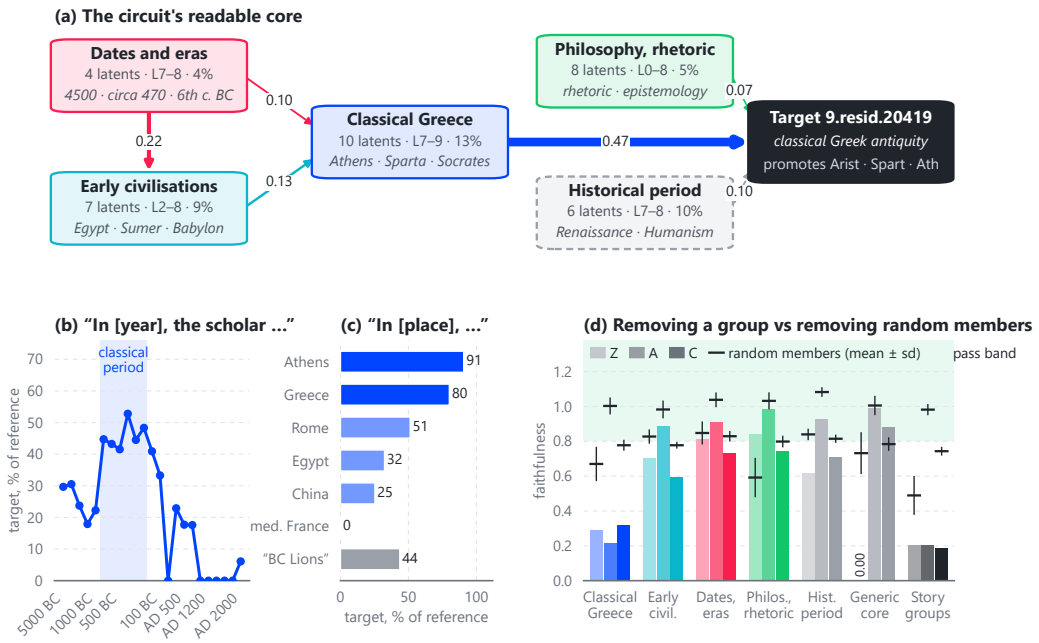


Figure 4: The weighted circuit of a latent for classical Greek antiquity (layer 9, residual stream; Section 5.3). **(a)** The readable core: role groups among the circuit's 60 largest contributors, named from their latents' own top contexts, with each group's size, layers, and share of the circuit's contribution at the target's anchor. Arrows give the drop in the destination when the source group is removed, under activating-mean ablation; the two left-hand groups also lower the target directly, by 0.11 and 0.12. The dashed grey group fires about as much on contrast contexts. Not drawn: a generic core of 7 latents and 682 further latents, together 55% of the contribution. **(b,c)** The target in the unmodified model on template sentences, as a percentage of its mean activation on its held-out strongest contexts. **(d)** Faithfulness with each group removed (bars) against removing the same number of random members of similar contribution rank (mean ± sd over 8 draws).

The coefficients carry causal weight. Necessity is unchanged by re-weighting, and the coefficients matter on a test they were never fitted to: setting nodes to α times their clean values induces the target in held-out contrast contexts, most strongly at residual-stream targets (Appendix E; pre-v1 panel, [*pending: sufficiency to induce with and without α after the refit, DAN-16*]).

5.3 A Circuit for Classical Antiquity

The running example shows what a passing circuit contains. Its target, a layer-9 residual-stream latent, fires where a passage places its subject in classical Greek antiquity, and its logit effect promotes "Arist", "Spart", "Ath", and "Greek". It tracks era and place rather than a word: completing "In [year], the scholar lived in the city of", it reaches 41–53% of its typical activation for 800–200 BC, 18–31% for older dates, and at most 6% from AD 1000 on, and for places it gives ancient Athens 91%, Rome 51%, Egypt 32%, and medieval France 0% (Figure 4b,c).

Its weighted circuit holds 728 latents and passes (Z, A, C 0.98, 1.11, 0.90; necessity 1.00). Grouping its 60 largest contributors by their own top contexts gives a readable core (Figure 4a). Ten latents at layers 7–9 respond to classical Greece itself (Athens, Sparta, Socrates, Thales) and carry 13% of the circuit's contribution; removing them takes the target to 0.29, 0.22, and 0.32 under Z, A, and C, against 0.67, 1.00, and 0.78 when as many random members of similar contribution rank are removed (Figure 4d). Upstream, latents for dates and eras feed latents for early civilisations

Arm	n **(med)**	$\phi_{\mathrm{Z}}^{\mathrm{free}}$	$\phi_{\mathrm{A}}^{\mathrm{free}}$	ϕ_{ind}	ϕ_{sup}	**pass**
path attribution (abl)	591,120	0.99	0.99	1.15	1.00	—
path attribution (ctr)	591,152	0.99	0.99	1.15	1.00	—
restoration (rounds=sites)	560,077	0.99	0.97	1.14	1.00	—
path attr. (abl), top-n	369	1.08	0.00	0.61	1.00	0/21
path attr. (ctr), top-n	369	0.99	0.00	0.56	1.00	0/21
restoration, top-n	369	4.36	0.00	0.66	0.95	0/21
hierarchical attr. (adapted)	359	0.63	0.00	0.86	1.00	0/21
unweighted circuit masking	1,216	0.66	0.23	0.75	1.00	3/21
weighted circuit, 100 steps	1,061	0.96	1.00	1.01	1.00	12/21
weighted circuit, 400 steps	369	0.99	1.08	0.89	1.00	**17/21**

Table 2: The cross-method matrix (pre-v1: fitted on 36 contexts and scored before the evaluation fixes of Appendix B; to be re-scored after the refit): medians over 22 depth-stratified targets, every score on held-out contexts; weighted-circuit rows apply their scaling coefficients (the α rows of Table 1). "pass" counts targets with faithfulness under both zero and mean ablation in $[0.8, 1.25]$ and $\phi_{\mathrm{sup}} > 0.9$, a provisional rule that predates the pass criterion of Section 4.4; one target's mean-ablation denominator is vacuous and excluded. Top-n rows truncate each attribution ranking to the target's weighted-circuit size, ranking each latent by its largest score at any context and position (the aggregate ranking of prior work is [*pending: the aggregate-ranking rerun in the v1 comparison suite*]); the hierarchical-attribution row applies the gating algorithm of Ge et al. (2024) to our SAE sites. Without transcoders our graph is not linear, so its scores are first-order gradients rather than their exact linear attributions, aggregated over contexts rather than per prompt, with the threshold bisected to the weighted circuit's size.

(Egypt, Sumer, Babylon), and both feed the classical group: removing the dates group lowers the early-civilisation group by 22% and the classical group by 10%. Latents for philosophy and rhetoric, the subject matter of the target's training contexts, matter little. Removing all 21 latents of the three story groups, 3% of the circuit, takes the target to 0.20, 0.20, and 0.19 (random: 0.49, 0.98, 0.74).

The core is a reading, not the whole circuit: the story groups carry 27% of the contribution and 682 further latents most of the rest, and "BC" alone partly triggers the target ("The BC Lions" reaches 44%). The circuit is also flagged as an amplifier under activating-mean ablation, restoring co-firing latents almost as well as the target, although it lifts only 104 latents above the Top-K cut against a run median of 578 under the same ablation (Appendix H). Five further cases, including a latent for names that carry a physical law, are in Appendix I.

5.4 Method Comparison at Matched Size

Table 2 compares the discovery families on the same 22 depth-stratified targets, the same held-out contexts, and the same ablation methods: the attribution baselines (Section 4.2) at their natural size and truncated to the weighted circuit's size, an adaptation of the hierarchical attribution of Ge et al. (2024), and unweighted and weighted circuit masking. [*pending: table re-scored after the protocol-v1 refit, DAN-16*]

Three results structure the comparison. First, at natural size every attribution arm reconstructs the target with its full per-position union, 10^5–10^6 nodes (the smallest faithful prefix of the same ranking is the 10^4–10^5 of Section 5.2); at that scale the arms' node sets largely coincide, so method identity matters only under compactness pressure. Second, truncated to the weighted circuits' size, every attribution arm fails, including the adaptation of Ge et al. (2024), whose threshold is bisected per target to matched size; unweighted circuit masking rarely passes; and the weighted circuit passes on most targets. Compact faithfulness is reached here only by optimising the node set *and* its coefficients together; the truncations are naive, and whether a better ranking narrows a gap from 0.00 to ∼1 under mean ablation at matched size is [*pending: the aggregate-ranking rerun*]. Third, a method wins the scores whose ablation method it was trained or ranked under, which we call the *home-turf effect* and

observed repeatedly across this programme, and under the prior work's own first-order evaluation the ranking inverts (Appendix E); rankings are therefore a matrix of discovery method against ablation method, never a single column.

5.5 Secondary Findings

Contrast contexts are a controlled variable. Semantic closeness and clean causal contrast are not the same property: for the contrastive attribution baseline, random contrast contexts beat the store's unchecked close ones on sufficiency to induce, while WCM circuits refitted under close, random, or distant silence-checked contexts move about as far as an identical refit does (Appendix G); [*pending: train-source × evaluation-source matrix at 288 targets, DAN-64*].

Discovered dependencies are not co-activation. Across 7,525 accepted attribution circuits, a circuit's nodes overlap its target's top co-activation partners by ∼3.6%, and correlational proposals fail the intervention gates outright; even granted fitted coefficients, a co-activation selection passes single scores but never the joint suite (Appendix F).

Output-targeted circuits do not explain latents, and vice versa. A reimplementation of the sparse-feature-circuits recipe (Marks et al., 2025) (integrated gradients one submodule at a time from a zero patch, error nodes, an output metric) on the same model and SAE bank scores $\phi_{\text{ind}} = \phi_{\text{sup}} = 0.000$ on each of the three deep target latents it was run on (layers 8–10), even at its most permissive threshold, and our circuits score near zero on logit faithfulness (Appendix M.1): "which latents support this output?" and "what makes this latent fire?" are answered by circuits that fail each other's objective. The output-targeted arm ran at its ceiling threshold, in-sample, and before protocol v1 and its evaluation fixes, so this is a scope result on three targets, not a ranking; [*pending: re-run under protocol v1, DAN-48*].

A feature relayed through the residual stream. Wiring the weighted circuit of a layer-11 MLP latent that fires on "p*iv*otal" with direct-effect edges (Appendix O.4) shows the residual stream carrying one feature forward: every upstream residual dictionary's best detector of the subword "iv" is a node, and edges shared by all three ablation methods link each layer's copy to the next. The copies are one feature: consecutive decoder directions are nearest or second-nearest neighbours among 40,960 latents (cosine 0.38–0.68) and co-fire on random text ($r = 0.87$–0.95).

5.6 Node Sets Are Non-Identifiable; Function Is Not

The masking engine is bit-deterministic at fixed settings (two identical runs agree to six decimal places with *zero* node flips), so tiny perturbations are controlled experiments on the objective, and every one we tried gives the same verdict (Appendix K). Float precision, the sparsity penalty, the stopping step, the training budget, and the contexts behind the mean-ablation values all leave size and faithfulness unchanged but exchange nodes: two node sets for the same target share only 40–85% of their latents. A discovered node set is one sample from an equivalence class of near-optimal solutions to (P), so we report node sets as equivalence classes and let set-level and functional claims carry the conclusions.

6 Discussion and Limitations

Circuit size is a measurement at a stated scaling convention. At natural scale a latent's circuit is large for deep targets (Section 5.2), and every one of those latents is individually removable at the noise level while bulk removal craters the target (Appendix D). A latent's activation, reached through an additive residual stream, is a dense function of many small contributions, so any rule that thresholds individual contributions undercounts it however the threshold is tuned. Fitted coefficients show that most of that collective tail is redundancy that fewer, rescaled latents can carry; a circuit's

size is therefore meaningful only alongside whether its coefficients were fitted, and weighted and unweighted sizes are compared only at matched faithfulness (Figure 3).

Ablation methods are part of the method. Miller et al. (2024) showed that faithfulness is not robust to the choice of ablation; our measurements push the point into training. The home-turf effect (Section 5.4), node sets trained under different ablation methods sharing fewer latents than node sets trained for different objectives, and the miscalibration of coefficients trained without zero ablation (Section 4.3) all say the same thing: the ablation methods used for discovery, training, and evaluation belong in the method statement. Further discussion (equivalence classes, position handling, composing circuits into chains) is in Appendix K.

6.1 Limitations

The SAE basis is not ground truth. Many latents resist clean characterisation, and circuits built from ambiguous latents inherit that ambiguity; our scores measure intervention effects, not interpretability.

The objective is a surrogate. A high pre-activation does not guarantee a Top-K slot, so WCM trains on a continuous surrogate of a discrete event; the rank-keep term narrows the gap on training contexts but does not guarantee the target's rank on held-out ones, and evaluation closes the loop by scoring the activation itself. For targets that barely enter the Top-K in the unmodified model the gap cannot be closed, which is why they are reported separately.

Faithfulness does not imply specificity. A circuit can restore its target by lifting the target's whole neighbourhood; the rank-keep term limits the lifts that overtake the target, but does not reliably remove such circuits, so specificity is measured and reported rather than claimed.

Wiring is descriptive and depends on the ablation method. A weighted circuit is fitted as a set of upstream latents sufficient for one target; direct-effect edges wire it (Appendix O.4), but the wiring is dense, much of it cancels, and edge sets agree only partly across ablation methods, so only edges shared by all three are interpreted, and composing circuits into chains still needs its own validation.

Contrast contexts. Contrast contexts are verified silent only in the sense that the target stays out of its Top-K; a near-miss pre-activation can remain, and closeness is measured in one pooled representation space, so "close" is one of several defensible constructions.

Generality. TuringLLM's synthetic corpus is an asset for control but a limit on generality, which the two public models partly address; larger models and other dictionary architectures remain open. The method is scoped to Top-K dictionaries by construction: on dense ReLU dictionaries, a random-circuit baseline with fitted coefficients loses its discriminating power once a set is large enough to span the anchors, so that architecture needs its own baseline design.

7 Conclusion

We reframed circuit discovery around a target latent, asking not which latents cause an output but what makes a latent fire, and answered with weighted circuit masking. At a sparsity penalty that yields circuits of 10^2–10^3 upstream latents, 46% of ∼15,000 sampled TuringLLM latents pass a strict held-out test, weaker penalties give larger circuits that pass more often, and without coefficients deep circuits are about 10× larger and pass half as often. Circuits for single latents are a building block: composing them into chains, with propagation tested at every link, is the path from dependencies to a map of the model's internal computation.

Reproducibility

The full system (pipeline stages, evidence stores, the discovery methods of Section 4, the target-intervention evaluation, and the analysis code that produces every figure and table in this paper) is available at `github.com/DanielJamesDavies/turing-explorer-circuit-discovery`. TuringLLM, its synthetic training corpus, and the trained SAE bank are described in Appendix A; every

experiment runs from a released configuration under the protocol of Appendix B, and each analysis reads only the run artifacts written by the pipeline stages.

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Attribute	Value
Parameters	254M
Layers	12
Residual stream dimension	1,024
Attention heads	16
MLP hidden dimension	4,096
Context length	1,024 tokens
Vocabulary size	50,304 tokens
MLP activation	SwiGLU
Normalisation	RMSNorm
Tokenizer	Phi-3 Mini

Table 3: TuringLLM-1.0-254M architecture summary.

A Corpus and Model Details

A.1 A Corpus Built by Design

The synthetic corpus was constructed in two stages. The first produced a relatively small but high-quality foundation: Phi-3 Mini (Abdin et al., 2024), a 3.8B-parameter language model available under a permissive licence, generated instructional and educational text spanning 18 broad subject areas (Physics, Mathematics, Computer Science, Chemistry, Biology, Philosophy, English, Creative Writing, Languages, Art, Music, Engineering, Economics, Business, Geography, History, Culinary Arts, and Communication), recursively generating topics and sub-topics and producing structured documents of headings and text.

The second stage addressed scale. A fine-tuned T5-based paraphrase model (Vorobev and Kuznetsov, 2023; Raffel et al., 2020) expanded the corpus by generating 11 paraphrased variants of each source document, motivated by knowledge distillation results (Ba and Caruana, 2014; Hinton et al., 2015) suggesting that varied outputs of a stronger teacher can be advantageous over raw data alone. The final corpus totals approximately 2 billion tokens.

Two tokenized datasets were derived using the Phi-3 Mini tokenizer. The language-modelling training set combines original and augmented texts shuffled into epochs, concluding with a second pass over only the high-quality originals to consolidate representations around the cleaner signal. The interpretability dataset consists of 64-token chunks shuffled into 4,096 shards, allowing the pipeline to process one shard at a time with a bounded memory footprint.

A.2 Architecture and Training

TuringLLM follows the residual-stream design of the GPT family (Radford et al., 2019) with two deliberate departures: SwiGLU (Shazeer, 2020; Ramachandran et al., 2017) replaces the ReLU MLP, following the motivation of the Llama family (Touvron et al., 2023), and RMSNorm (Zhang and Sennrich, 2019) replaces layer normalisation. Table 3 lists its dimensions and Figure 5 sketches the architecture. Training reached a minimum cross-entropy of approximately 1.62 (train) and 1.80 (validation); the same infrastructure applied to GPT-2 (124M) reaches approximately 3.29 (Karpathy, 2024), reflecting the tighter distribution of the synthetic corpus. The loss curve (Figure 6) drops sharply in its final phase, corresponding to the second pass over Phi-3 Mini originals.

A.3 Fitness for Interpretability

TuringLLM is not designed to win benchmarks; it is designed to be a controlled subject. Standard held-out benchmarks fit poorly for a base model trained on synthetic educational text, so fitness was evaluated qualitatively and through structured internal tests: Phi-3 Mini composed topic-specific opening sentences and scored TuringLLM's completions on English quality and topic adherence,

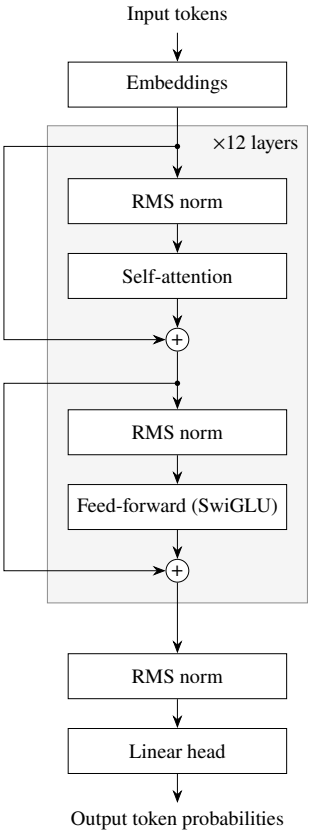


Figure 5: TuringLLM-1.0-254M architecture: a 12-layer pre-norm decoder-only transformer with RMSNorm and a SwiGLU feed-forward block. The circuit-discovery pipeline attaches SAEs at three sites per layer: attention output, MLP output, and the residual stream.

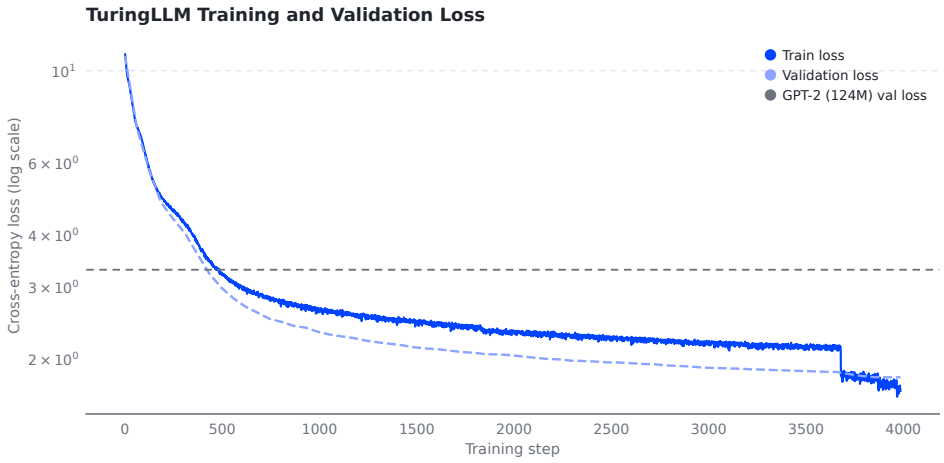


Figure 6: TuringLLM training and validation cross-entropy over the 3,985-step run (log scale). The dashed reference marks the validation loss reached by the same training infrastructure applied to GPT-2 (124M); the drop near step 3,700 corresponds to the final pass over only the high-quality Phi-3 Mini originals.

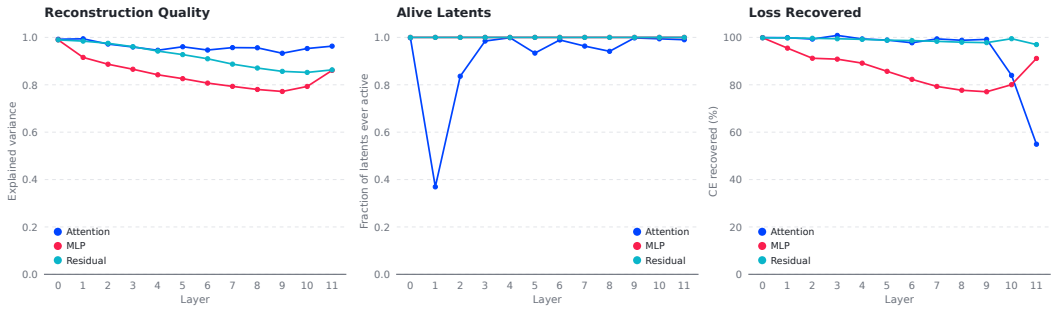


Figure 7: Quality of the 36-SAE bank by site. Left: explained variance of each SAE's reconstruction. Middle: fraction of latents that fired at least once over the full collection pass. Right: percentage of the cross-entropy gap recovered when splicing each SAE's reconstruction into the model, relative to zero-ablating the site.

yielding average grades of approximately B and A respectively across the 18 subject areas. This level of competence supports the working hypothesis that the model contains structured internal representations worth studying; a model merely interpolating noise would not produce consistently topical completions. Representative completions are shown in Figure 8.

A.4 Sparse Autoencoder Quality

Figure 7 reports three standard quality metrics for the 36-SAE bank, per site. *Explained variance*, $1-\mathrm{Var}(x-\hat{x})/\mathrm{Var}(x)$, measures reconstruction fidelity. *Alive latents* is the fraction of each dictionary's 40,960 latents that fired at least once during the full collection pass over the interpretability corpus. *Loss recovered* splices each SAE's reconstruction into the model at its site and reports the recovered share of the cross-entropy gap between the original model and zero-ablating that site (Marks et al., 2025), measured over 12,288 tokens of the interpretability corpus. Because the SAEs are Top-K with $K = 128$, sparsity is fixed by construction and the usual L_0–fidelity trade-off curve is a single point per SAE.

Three observations matter for interpreting the paper's results. First, reconstruction fidelity is healthy across the bank: attention SAEs explain 0.93–0.99 of variance, residual SAEs decline smoothly with depth to 0.85, and MLP SAEs bottom out at 0.77 in layer 9. Second, the layer-1 attention SAE has a dead-latent crater: only 37% of its latents ever fire (84% at layer 2, at least 93% everywhere else). Third, loss recovered exposes a functional gap that explained variance misses: the layer-11 attention SAE recovers only 55% of the cross-entropy gap despite explaining 0.96 of variance (layer 10: 84%), indicating that the unexplained residual at the final attention site is disproportionately important to the logits.

B Protocol

Every circuit in this paper is discovered and scored under the settings in Table 4, on every model; only the model and its dictionary change.

C Attribution Baselines and Position-Aware Node Sets

C.1 Attribution Objectives

Two gradient objectives supply the signed scores, one for each side of the target's behaviour.

"Physics utilizes the fundamental concepts of quantum mechanics to explain how electrons populate a sublevel within an atom, which is marked by energy levels. These levels define the overall potential energy and enable electrons to occupy specific energy levels or sublevels [. . .]"

"Artificial Intelligence (AI) has played a pivotal role in elevating personalization in human-robot interfaces, elevating the potential for truly personalized and responsive user experiences. The evolution towards AI extends its roots back to the 1960s with early rule-based [. . .]"

"Artificial Intelligence (AI) and Machine Learning (ML) technology have become indispensable in modern engineering. These technologies enable engineers to analyze vast datasets with unprecedented efficiency, leading to significant improvements in design processes, maintenance, and optimization."

Figure 8: Representative TuringLLM completions (excerpts). Continuations are topical and grammatical, supporting the use of the model as an interpretability subject; occasional repetitions and factual slips (e.g. conflated terminology) reflect its 254M-parameter scale and synthetic corpus.

Contrastive gradient: why does the target not fire? When a latent is absent from the Top-K code its activation is identically zero and a gradient through it carries no information, so attribution targets the target latent's *encoder pre-activation*

$$\hat{a}_f(x,t) = \left\langle W_e^{(f)},\, z(x,t) \right\rangle + b_e^{(f)},$$

with $W_e^{(f)}$ the target's encoder row, $b_e^{(f)}$ its effective bias, and $z(x,t)$ the pre-SAE dense activation. The pre-activation is defined on every forward pass whether or not f clears Top-K, and is differentiable with respect to all upstream latents through identity passthroughs at each SAE site. On a contrast context x, the objective is the squared gap $\mathcal{L}^{\text{ctr}}(x) = \left(\hat{a}_f(x,t^*) - \mu_f^+\right)^2$ to the target's mean pre-activation over activating contexts, taken at the position of maximum pre-activation, and each upstream latent g present in a contrast context is scored by gradient-times-activation averaged over contrast contexts: $S_g^{\text{ctr}} = \frac{1}{|\mathcal{N}_f|} \sum_x a_g(x,t_g)\, \partial\mathcal{L}^{\text{ctr}}(x)/\partial a_g(x,t_g)$. A latent absent from a contrast context has $a_g = 0$ there, so it is scored by the raw gradient $\partial\mathcal{L}^{\text{ctr}}/\partial a_g$ instead: a sensitivity score for latents the circuit would have to switch on. The baseline keeps the top-scoring latents of each sign of S_g^{ctr} (Algorithm 1). The source of contrast contexts is a controlled variable: `close` (the retrieved store), `random`, or `distant`.

Ablation gradient: what sustains the firing? The mirror score on activating contexts, $S_g^{\text{ab}} = \frac{1}{|\mathcal{P}_f|} \sum_{(x,t)} a_g(x,t_g)\, \partial\hat{a}_f(x,t)^2/\partial a_g(x,t_g)$, is large when the upstream latent is strongly active and the target is sensitive to it; its top latents form the node set. The `local` mode differentiates the squared pre-activation, which weights each context by twice the target's pre-activation there; the path modes differentiate $\hat{a}_f$ itself.

Hybrid fusion. A hybrid arm runs both objectives on the same target and fuses the circuits by latent identity, each node keeping its per-source scores; fusion details, pruning, and acceptance modes are in Appendix L and Algorithm 3.

Co-activation graph. A second pass over stored top contexts computes, for every latent, its top 64 co-activating partners scored by clamped pointwise mutual information. In this paper it serves as the correlational reference against which the causal discoveries are compared (Appendix F).

C.2 Attribution Modes

Both objectives are estimator-agnostic: evaluating the gradient at the clean activations is one choice of *linearisation point*, not the only one. Four modes differ only in where gradients are linearised, holding objective, probes, and selection fixed:

- **`local`**: a single linearisation at the natural activations (gradient-times-activation at the clean forward pass). This is attribution patching (Nanda, 2023; Syed et al., 2023) with a zero patch and the gradient taken on the clean run, the AtP form of Kramár et al. (2024); Nanda (2023) takes the gradient on the corrupted run.
- **`ig_mean`**: integrated gradients along one straight path from every upstream site's mean-ablation value to its natural activations, all sites interpolated together, with downstream sites held at their interpolated codes and the SAE errors at their clean values. It follows the $\mathrm{IE}_{\mathrm{ig}}$ node estimator of Marks et al. (2025) in form but not in detail: theirs interpolates one submodule at a time with the rest of the model live, starts from a zero patch when no counterfactual exists, and scores the error terms as nodes.
- **`restoration`**: iterated greedy re-linearisation, in which each round attributes at the *current* partially restored stream, admits the top scorers, restores them by live re-encoding, and re-linearises, for as many rounds as the target has upstream sites (Algorithm 4).
- **`ig_restoration`**: the restoration loop with per-round integrated-gradient scoring.

The taxonomy is single-point, path, and iterated-trajectory linearisation. A path mode makes the attribution baselines comparable in kind to the published estimator, but not identical to it, so the output-targeted comparison of Section 5.5 uses its own reimplementation of the published recipe; a version of that recipe with the target latent as its metric, which would isolate the choice of target from everything else, is [*pending: DAN-48*].

C.3 Node Sets and Position-Aware Discovery

Attribution returns ranked latents, so the baselines must also decide which node set a ranking defines. The Top-K constraint makes the classical answer fail structurally: the code is k-sparse *per position* while a target reads its whole causal prefix, so a position-collapsed circuit must cover the union of every position's active set, thousands of distinct latents per site, or recomputed faithfulness collapses. The failure is a property of position aggregation, not of node selection (mechanism measured in Appendix N.1).

The baselines' node set is therefore a set of latents, the same kind of object (P) optimises over, collected across positions. Latents outside the circuit are ablated everywhere; nodes are not forced to any value: they are recomputed from the modified stream at every position, so positions instantiate naturally (allow, don't force), exactly the circuit-only execution of Section 4.1. Discovery is position-aware to match: attribution is computed per position, a percentile cut on |attribution| (pooled over sites and positions, resolved per batch of four contexts) keeps individual (latent, position) entries, and the node set is the union over the positions of the target's causal prefix. With a mean-ablation path nearly every entry is nonzero, so the median cut keeps about half of them, which is why these unions reach 10^5–10^6 nodes (Table 2). No template or cross-example alignment is required, and in our sweeps the percentile rule dominates top-N and per-position–normalised alternatives. Both attribution signs are kept, as a design principle rather than an implementation detail: the node set is *stream reconstruction*, not a selection of what drives the target, and a negative-attribution latent still carries part of the residual stream that nodes are recomputed from, so removing it corrupts recomputed execution.

Two optional, validated levers shrink the set: a *recurrence prune* (nodes must fire in at least two contexts) and a *magnitude-bisection prune* (a binary search over the attribution ranking for the smallest prefix holding a requested faithfulness, automating the threshold-tuning step of prior practice Marks et al., 2025); details and their depth-dependence are in Appendix O.

A node set's size at a stated faithfulness is a measured quantity, and Appendix D argues it measures the structure of the computation rather than a defect of the method.

D Control versus Faithfulness at Natural Scale

Which latents can impose a target's activation? A few hundred. How many must remain for the circuit alone to reproduce it? Tens of thousands. The two questions are two criteria over the

same node set. *Interventional control* asks for a small set whose intervention scores clear an acceptance gate, sufficiency to induce $\phi_{\text{ind}} \geq \tau_{\text{ind}}$ and necessity $\phi_{\text{sup}} \geq \tau_{\text{sup}}$, as the gradient attribution methods of Appendix L used. *Faithfulness* is (P) with every coefficient fixed at $\alpha = 1$: the smallest node set that reproduces the target alone, whose size n_ε we call the target's *faithfulness cost*. The clamped/recomputed decomposition separates them, and on depth-spanning target panels their answers come apart; this appendix measures how far, for node sets from the attribution baselines of Appendix C. All numbers here come from panels of three to four targets per layer under the evaluation that preceded protocol v1, and are indicative.

Control is concentrated. Restricting a node set to its most influential positions barely moves clamped faithfulness: it reaches 0.79–1.04 with only 10% of positions kept, roughly flat as more are added. What *causally controls* a target lives in a handful of positions and a compact set of latents: a circuit in the classic, hand-annotatable sense. For a pre-v1 thermodynamics latent (layer 3), the controlling circuit reads like a chemistry syllabus: latents for Arrhenius kinetics, the third law, and absolute zero.

Faithfulness is collective, position-local, and depth-scaled. Recomputed faithfulness tells the opposite story. At the same 10% of positions a shallow target still passes (0.85), but deep targets sit at exactly 0.00 and climb only position by position as the rest of the prefix is restored (a layer-7 residual-stream target: $0 \to 0.24 \to 0.52 \to 0.82$). This is the sense in which the cost is *position-local*: no small set of positions carries it, and the score accrues position by position across the causal prefix. The faithfulness-versus-size curve (Figure 9) is two-regime on every target we measured: most compression is available down to $\phi^{\text{free}} \approx 0.7$, then a hard limit, an irreducible core that even $\phi^{\text{free}} = 0.2$ requires. That core is the faithfulness cost n_ε at loose tolerance, and it scales with depth: roughly 2k at layer 3, 10k at layer 7, 60k at layer 11. Compressibility follows the same gradient: shallow node sets prune to 13–23% of their size, deep ones only to ∼70% with a sharp cliff. Depth, not the selection rule, is the fundamental determinant of circuit size. These are costs *at natural scale* ($\alpha = 1$): Section 5.2 shows that fitted per-node scaling coefficients collapse them by a factor of ten to a hundred.

Individually negligible, collectively load-bearing. Leave-one-out is blind to all of this: removing any single node changes the target by 0.000, the noise level. Yet cutting nodes in bulk by their own attribution ranking craters function: on recurrence- and bisection-validated node sets, dropping the bottom half by score moves ϕ_Z^{free} from 0.975 to 0.895 at layer 2 but from 1.119 to 0.595 at layer 8; dropping the bottom 90% yields 0.635, 0.000, 0.222, and 0.005 at layers 2, 8, 9, and 10. The scores being cut are minuscule, $2\text{–}5 \times 10^{-4}$ in attribution units. Nodes individually at the noise level collectively carry the entire deep reconstruction. Individual-contribution thresholding, the standard selection rule across circuit discovery, therefore systematically undercounts what faithfulness needs; this is the finding that makes circuit size a measurement rather than a defect.

The union survives its controls. Node sets are target-specific, not noise or soup (cross-target node-set tails overlap at chance, Jaccard ∼0.07, against 0.91 for a target's own split halves); the plain magnitude union Pareto-dominates every structured alternative we tried; and at depth, faithfulness is partly carried by the preserved SAE errors, the latent-targeted echo of the error-node finding for output targets (Marks et al., 2025). The full controls are in Appendix O.

E Weighted Circuits: Sufficiency to Induce and the Training Budget

Weighted against unweighted circuits across the sparsity penalty. The comparison at a fixed penalty (Figure 10, left) uses the one value both grids share, $\lambda = 10^{-3}$; the frontier and the natural-scale cost come from the full sweep, which fits WCM at $\lambda \in \{2.5, 5, 10, 20, 40\} \times 10^{-4}$ and unweighted

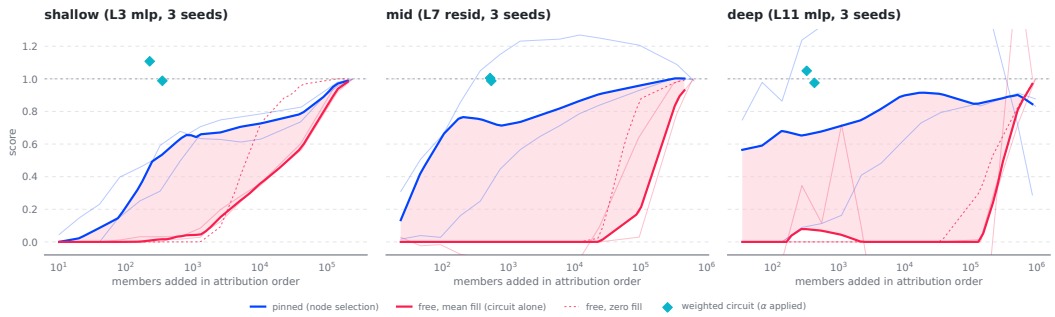


Figure 9: Control is sparse; faithfulness is collective; scaling coefficients collapse its cost. Cumulative score against nodes added in attribution order (three targets per depth band: median curves, individual targets ghosted; position-aware path attribution, percentile selection). Clamped faithfulness (node selection) rises steeply within 10^2–10^3 nodes, while recomputed faithfulness (the circuit alone, at $\alpha = 1$) sits near zero until 10^4–10^5 nodes, under mean ablation (solid) and zero ablation (dashed). The shaded clamped–recomputed gap widens with target depth. Diamonds: *weighted circuits* discovered on the same targets (Section 5.2), scored with their scaling coefficients applied under zero ablation, a few hundred nodes reaching the score the $\alpha = 1$ curve needs 10^5 for.

circuit masking at $\lambda \in \{0.1, 0.3, 1, 3, 10\} \times 10^{-4}$ to 191 targets (five per layer and site kind, plus the pilot targets; 190 with usable contexts) and scores every circuit on the held-out strongest contexts. Figure 3 adds WCM at $\lambda \in \{0.25, 0.5, 1\} \times 10^{-4}$ for every sweep target, so that both methods reach circuits of $\sim 10^4$ latents and beyond.

Unweighted masking needs a weaker penalty to reach comparable faithfulness, so the two grids differ. Figure 15 separates the three ablation methods: the gap is smallest under zero ablation and largest under the two mean ablations, where unweighted circuits of $\sim 10^3$ latents score 0.18–0.37 at layers 5–11 (zero ablation 0.58–0.72). For the natural-scale cost we take, per target, the smallest circuit whose worst score lies within 0.3 of 1. Among targets both methods reach, the medians are 128 against 502 latents at layers 0–4 (3.7×; 66 of 76 targets reached by both), 447 against 3,862 at 5–7 (9.8×; 33 of 48), and 789 against 9,989 at 8–11 (14.1×; 25 of 66). On this grid WCM stops near 10^3 latents, so at depth it reaches that looser criterion for fewer targets (31 of 66) than unweighted masking does at up to 3×10^4 latents (39 of 66); the three weaker WCM penalties of Figure 3 remove that limit. The same holds for the pass criterion itself: taking, per target, the smallest circuit that passes on the five-penalty grid, the medians are 123 against 514 latents at layers 0–4 (3.8×; 60 targets pass under both methods), 380 against 5,860 at 5–7 (11.5×; 25), and 672 against 10,223 at 8–11 (14.1×; 20). With the three weaker WCM penalties more targets pass under both methods, and the ratios are 3.7× (62 targets), 9.6× (34), and 10.5× (26; 906 against 13,075 latents), the values quoted in the body. Pooled over depth bands, WCM's pass rate rises monotonically as the penalty weakens: 31%, 35%, 42%, 47%, and 58% at $\lambda = 40, 20, 10, 5, 2.5 \times 10^{-4}$, where the median circuit holds 218, 327, 472, 747, and 1,103 latents; the production setting is one point on this curve, which has not levelled off. Across penalties (Figure 13) the two methods meet at layers 0–4, where both pass ~78% of targets (WCM at 616 latents, unweighted at 3,730); at 5–7 WCM passes 56% at 1,176 latents where unweighted masking needs 9,037 for 58%, and at 8–11 WCM passes 36% at 1,637 where unweighted masking needs 32,243 for 39%. At the production penalty the per-score medians are all lower unweighted but not dramatically so at layers 8–11 (zero ablation 0.58 against 0.89, activating mean 0.27 against 0.69, contrast mean 0.18 against 0.51; Figure 14); the pass rates differ far more because the criterion needs every score at once.

The coefficient triad. Figure 11 sets the three versions of a circuit side by side, each on its own stratified sample of production targets: the circuits as fitted (46% pass), the same nodes with every coefficient set to 1 (1.5% of 1,746; at most 4% in any depth band or site kind), and random nodes of

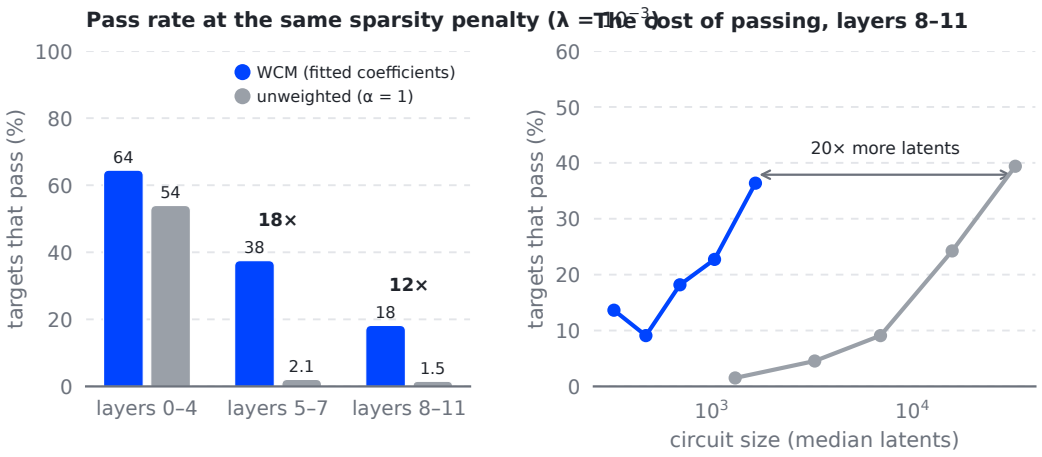


Figure 10: Weighted against unweighted circuit masking on the five-penalty grid. Left: the share of targets whose circuit passes the criterion of Section 4.4 (faithfulness under Z, A, and C in $[0.8, 1.5]$ and necessity ≥ 0.9, on held-out contexts) for weighted circuit masking (blue) and unweighted circuit masking ($\alpha = 1$, grey) at one sparsity penalty, $\lambda = 10^{-3}$ (the production setting), by target depth band; bold labels give the ratio. Right: pass rate against median circuit size across the sweep of sparsity penalties, layers 8–11; the arrow compares WCM's highest pass rate on this grid with the unweighted point of similar pass rate. 190 targets, protocol v1; every depth band is in Figure 13 and the per-score medians in Figure 14.

the same size given the same coefficient fitting (0 of 747). At natural scale the nodes seldom overshoot the target: 4 of the 1,746 circuits exceed 1.5 under any ablation method, all at attention-output targets. More often they fall short of the empty circuit itself under activating-mean ablation (18%; 7% at layers 0–4, 32% at 8–11, and 29% of attention-output targets). There the activating-context means that the members replace already carry more of the target (a median 1.16 of a natural 3.06) than the members do at their natural scale (0.61), and the fitted coefficients make up the difference; these circuits pass less often with fitted coefficients (16%, against 51% for the other targets).

What the coefficients do to members. Comparing each member's value inside the circuit-only run with its value in the unmodified model, at the target's anchors on held-out contexts, shows that the coefficients do not return members to their natural values (Figure 12a; 600 production circuits, 200 per depth band). Before the coefficients act, the ablated environment leaves 14% of members unchanged, pushes 8–10% out of their site's Top-K, and raises 11–14% to three or more times their natural value; 28–30% lie within 25% of it. After the coefficients fewer do (22–23%), with the spread widening on both sides, and weighting members by their gradient-times-activation contribution to the target gives the same picture. A member's coefficient is unrelated both to how far the ablation moved it and to how much it contributes (rank correlations within ±0.1). What the coefficients change is the target. With every coefficient set to 1, the members alone bring the target to a median 0%, 19%, and 7% of its natural activation under Z, A, and C (21%, 24%, and 12% of its pre-activation), against 94%, 97%, and 90% with the fitted coefficients (Figure 12b). To first order the coefficients raise the members' summed contribution only about 1.6×, so most of their effect on the target arises in the layers between the members and the target; we report this as an observation, not a mechanism. Deep and failing circuits also lose members to the ablated environment itself: weighted by contribution, 10% of members of layer 8–11 circuits are pushed out of Top-K under contrast-mean ablation, against 1% at layers 0–4, and 9% in failing circuits against 1% in passing ones.

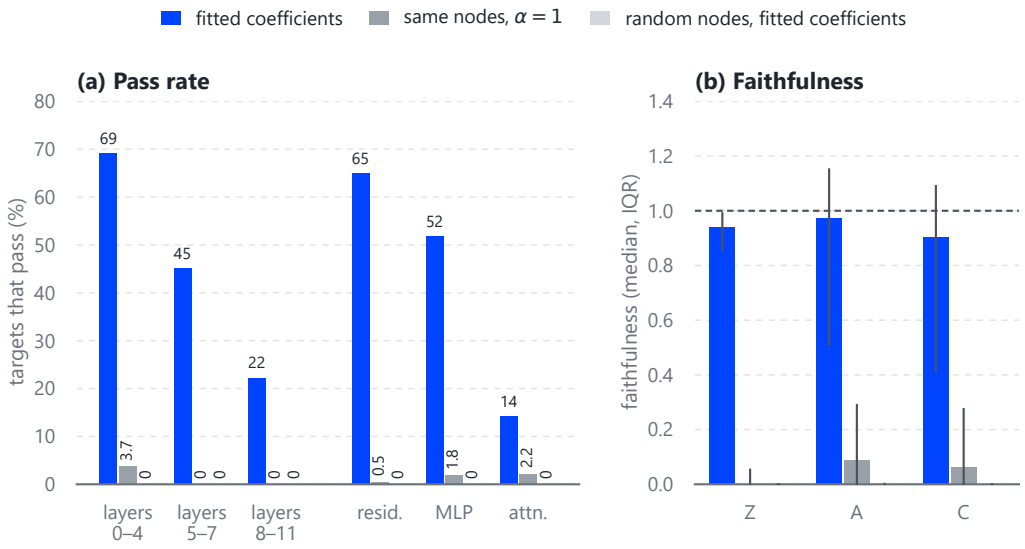


Figure 11: The coefficient triad on the production circuits. **(a)** The share of targets that pass the criterion of Section 4.4, by target depth band and site kind, for the circuits as fitted (blue; the full run, 14,994 circuits), the same nodes with every coefficient set to 1 (grey; 1,746 targets), and random nodes of the same size given the same coefficient fitting (light grey; 747 targets). **(b)** Median faithfulness under zero (Z), activating-mean (A), and contrast-mean (C) ablation, with interquartile ranges; 1 is perfect (dashed).

Sufficiency to induce by site kind. Sufficiency to induce, unlike reconstruction, is structured by the target's site kind (Figure 17). It asks a question no circuit is trained on: setting the circuit's latents to α times their mean values at the anchors of the activating contexts, does the target switch on in held-out contrast contexts where it is silent? Across the production run the median is 0.84 (0.98 among passing circuits, of which 88% switch the target on to at least half its natural level). Residual-stream targets are induced almost everywhere (96% at half or more; median 1.19, rising with depth to 1.35 at layers 8–11), MLP-output targets partly (70%; median 0.69), and attention-output targets rarely beyond layer 4 (28% and 29% at layers 5–7 and 8–11; median 0.20 overall). Which latents *can* be induced from upstream is therefore itself a per-kind property, and characterising it is open.

The budget is part of the method. Sweeping the training budget shows reconstruction faithfulness converges by 100 steps; everything after buys sparsity. That sparsity is paid for in the calibration of sufficiency to induce: on the panel, deep targets are induced with overshoot at the full budget (median $\phi^{\alpha}_{\text{ind}}$ 1.48) and calibrate at a quarter of it (1.10 deep, 1.01 shallow), at two to three times the circuit size. The sharpest case replicates held-out. A layer-9 circuit that reconstructs perfectly yet barely induces its target at the full budget (0.17) *induces* it (0.83) when discovered at a quarter of it; the calibration sweep shows the nodes that carry the induction being progressively pruned as training extends (0.88 → 0.57 → 0.25). Sparsity bought by training budget and sparsity bought by the penalty are also *different circuits*: at matched size the two prunings share only about half their nodes, a fifth non-identifiability probe (Section 5.6). Two further results on the training ablation methods close the loop. Coefficient-only training on a frozen node set does not fix the induction overshoot but does inflate the coefficients monotonically; the objective simply does not constrain sufficiency to induce. Removing the zero-ablation term from training explodes zero-ablation faithfulness to 36–1,681 at depth while the trained ablation method stays in band: the sharpest demonstration of the *home-turf effect* (Section 5.4), now at the level of coefficients rather than node sets. The training schedule therefore belongs in the method statement: the protocol fixes the full budget, and a quarter of

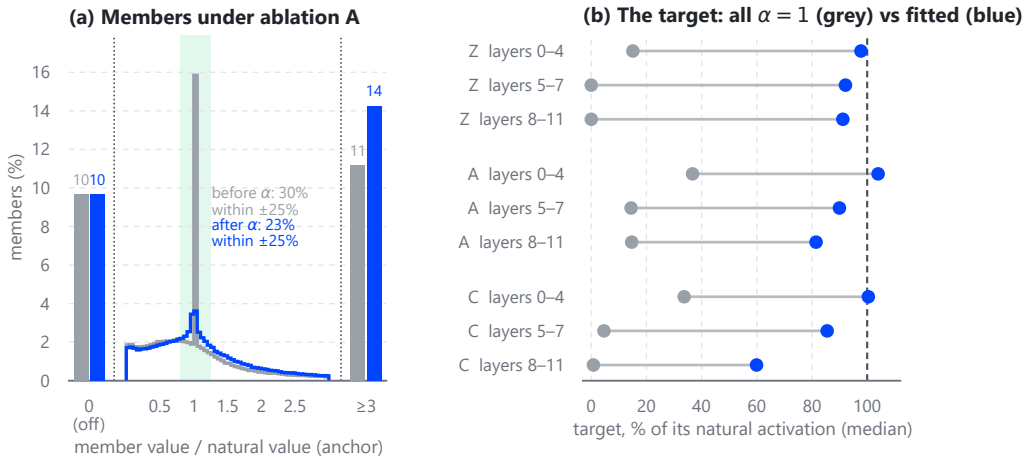


Figure 12: What the coefficients do to circuit members (600 production circuits). **(a)** Each member's value in the circuit-only run divided by its value in the unmodified model, at the target's anchors on held-out contexts, under activating-mean ablation, before (grey) and after (blue) its coefficient; members active in the unmodified model, pooled over circuits. The bars at either side collect members that left their site's Top-K ("0, off") and members at three or more times their natural value; both groups are mostly members that fire weakly in the unmodified model, and together they carry ~8% of the circuits' contribution to their targets. **(b)** The target's activation in the circuit-only run as a share of its natural activation, median over circuits, with every coefficient set to 1 (grey) and with the fitted coefficients (blue), by ablation method and target depth band.

it (with a stronger penalty) calibrates sufficiency to induce better at two to three times the circuit size.

Training dynamics at scale. Every production fit records its per-step curve (Figures 18 and 19). All gates start open, so no latent falls below the keep threshold for the first ~55 steps; membership then collapses from 10^5–10^6 candidates to a median of about 1,700 by step 100, and a circuit is within twice its final size by a median step of 137. It is not converged at the protocol's 400 steps: the median circuit loses 12% of its members over the last 100 steps (16–17% at layers 10 and 11), and 88% are still shrinking by more than 5%. The data loss stays nearly flat over that stretch while the penalty falls, so the late pruning costs little training fit, but at depth the circuits still shrinking pass less often: 30% against 67% for the settled half at layers 4–7, and 9% against 36% at layers 8–11, with no difference at layers 0–3. Reported sizes are therefore sizes at a fixed budget. The sparsity penalty dominates the loss throughout: its tail of nearly closed gates outweighs the members, and the data terms end at 0.3% of the loss. Before their 0.25 weights, the contrast-mean and activating-mean errors are about twice the zero-ablation error, the same ordering as held-out faithfulness.

Attribution-mass recovery. Ge et al. (2024) score circuits by attribution recovery inside their linearised graph, with no intervention; our first-order analogue is *attribution-mass recovery*, the share of total activation-times-gradient mass a node set captures (a surrogate: the exact leaf-sum identity exists only in their substitute model). Under it the ranking of Table 2 inverts: plain top-n attribution leads (median signed recovery 0.76), the replication follows (0.72), and the weighted circuit trails (0.52) while being the only arm that passes the intervention test. The home-turf effect therefore reaches evaluation frameworks themselves, and the gap carries a finding: a circuit that reproduces the target on held-out intervention while holding a third of the absolute attribution mass (0.37 median) shows that first-order attribution mass is not what faithfulness is made of.

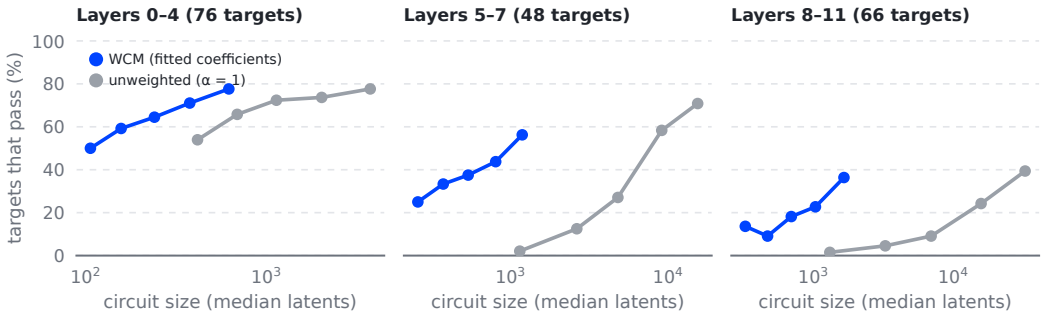


Figure 13: The pass-rate frontier for every target depth band: the share of targets that pass the criterion of Section 4.4 against median circuit size, one point per sparsity penalty, for weighted circuit masking (blue) and unweighted circuit masking ($\alpha = 1$, grey). 190 targets, protocol v1.

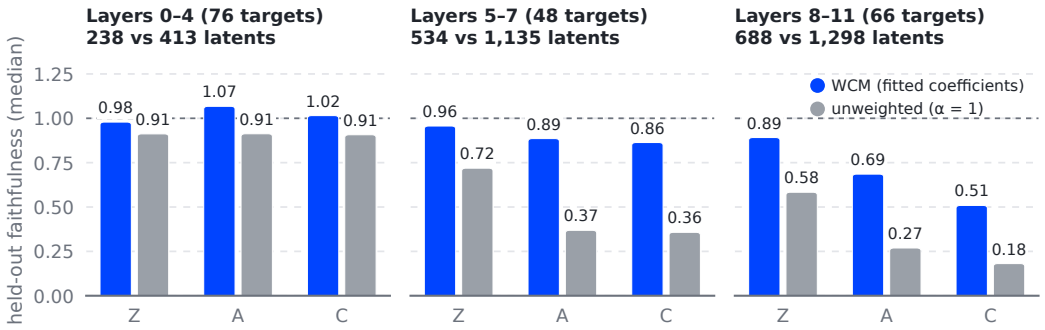


Figure 14: Per-score medians at the production sparsity penalty, $\lambda = 10^{-3}$: held-out faithfulness of weighted (blue) and unweighted (grey) circuit masking under zero ablation (Z) and mean ablation over activating (A) and contrast (C) contexts, by target depth band; panel titles give the median circuit sizes, weighted against unweighted. 1 is perfect (dashed).

F Dependencies Are Not Recoverable from Co-activation

If the discovered dependencies were merely co-occurrence in disguise, the co-activation graph would predict them. It does not. In a full-scale run of the gradient methods producing 7,525 accepted circuits, the mean overlap between a circuit's discovered nodes and its target's top co-activation neighbourhood was approximately 3.6%, the mean pairwise co-activation density among discovered nodes was approximately 0.26%, and neither quantity correlated with sufficiency to induce (Pearson $r = 0.02$; distributions in Figure 21). Latent co-activation profiles across the bank are themselves nearly orthogonal (the first two principal components explain roughly 3% of variance), so local co-occurrence structure carries little of the information that gradient-based contrast discovers. Causally validated dependency neighbourhoods, in short, are not local co-activation neighbourhoods.

The converse test is to let co-activation structure propose circuits directly. Under the same 128-target protocol, evidence base, and target-intervention gates, two correlational baselines produced no accepted circuits at all, while ablation gradient accepted circuits for 121 of the same 128 targets (Table 7). Sparse co-activation expansion follows the strongest co-activation edges outward from the target; differential activation ranks latents by their activation contrast between activating and contrast contexts. Correlational proposals do not merely score lower; they fail the intervention gates outright. The strongest version of this challenger — anchor co-activation selection granted the weighted circuit's own coefficient fitting — revives on reconstruction but fails the joint calibration

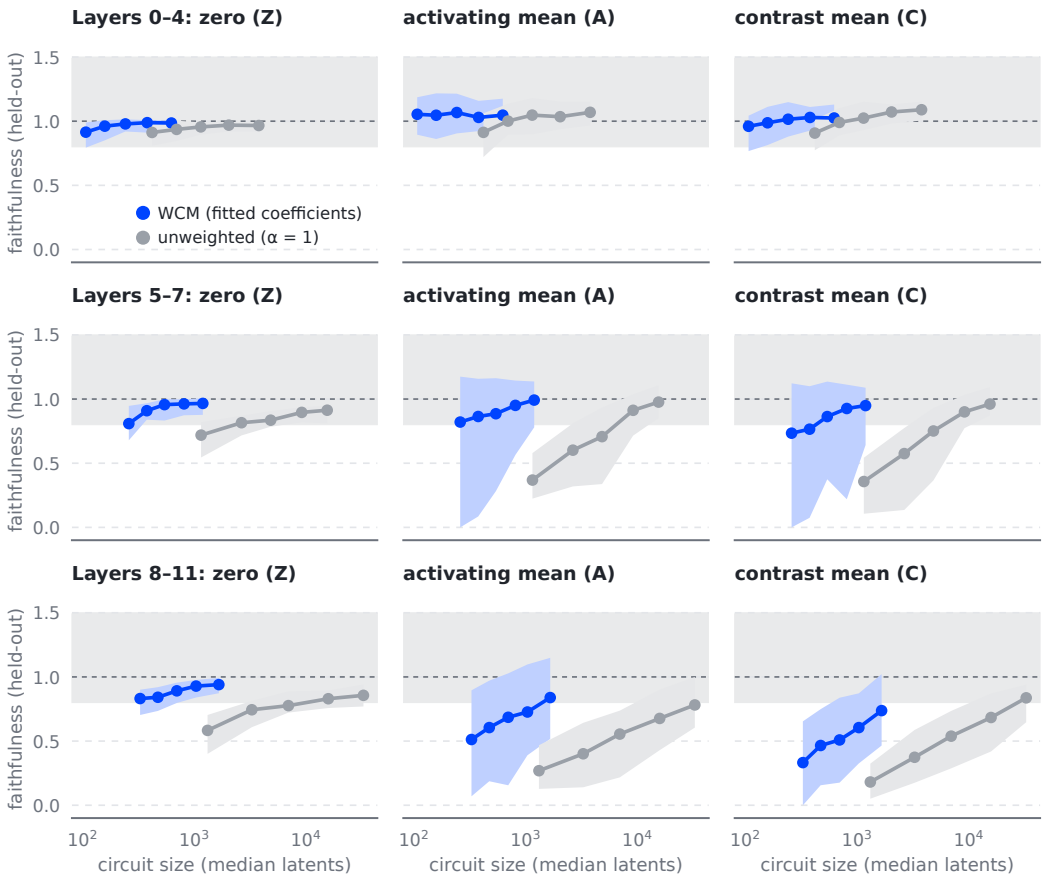


Figure 15: Weighted against unweighted circuit masking by ablation method (rows: target depth band; columns: zero ablation, and mean ablation over activating and over contrast contexts). Median held-out faithfulness and interquartile range against median circuit size, one point per sparsity penalty; the grey band is the pass band $[0.8, 1.5]$. The gap between the methods is smallest under zero ablation, largest under the two mean ablations, and widens with depth.

suite on every target tested, and is treated in Appendix Q.2.

G Method and Contrast-Context Source

On the attribution grid (Figure 20), contrastive gradient achieves the highest median sufficiency to induce across all three sources, peaking at approximately 0.90 with random contrast contexts. Its advantage over ablation gradient is largest in the random condition (0.36), intermediate in the distant condition (0.24), and smallest in the close condition (approximately 0.09). This ordering makes mechanistic sense: contrastive gradient is explicitly optimised for sufficiency to induce, while ablation gradient targets necessity and achieves sufficiency to induce as a by-product.

The effect of the contrast-context source is itself a finding. One might expect close contrast contexts, semantically similar to the activating contexts but lacking the target's activation, to produce the sharpest contrastive signal. In practice the reverse holds for contrastive gradient: random contrast contexts outperform close ones by approximately 0.16 median points of sufficiency to induce. The most likely explanation is that close contrast contexts, being near the activating distribution, partially activate the target's mechanism, providing a noisy gradient signal (this grid used the retrieval store, whose contrast contexts are not checked for silence; Appendix M), while random contrast

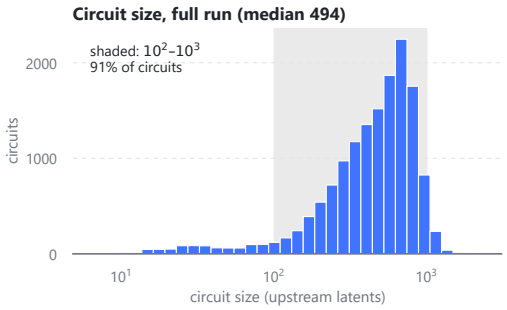


Figure 16: Circuit sizes in the production run at the production sparsity penalty: 91% of circuits hold 10^2–10^3 upstream latents (shaded); median 494.

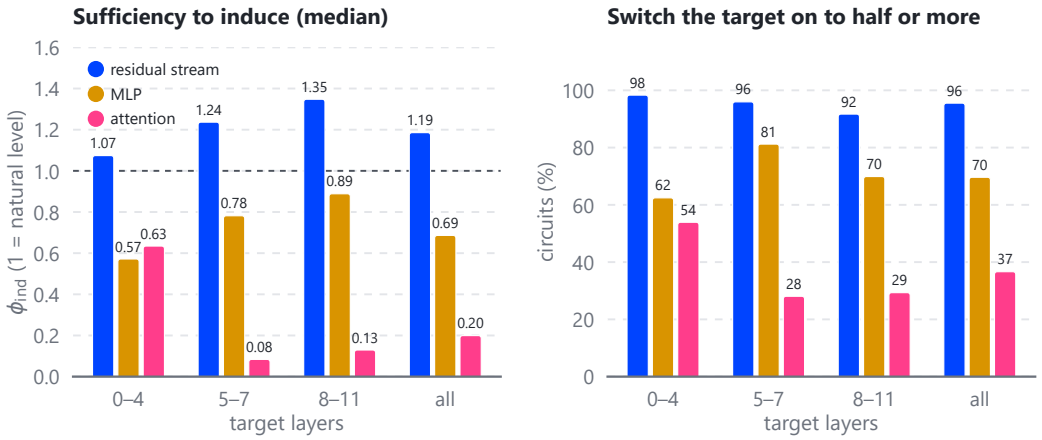


Figure 17: Sufficiency to induce across the production run (15,004 TuringLLM circuits), on held-out contrast contexts where the target is silent. Left: median $\phi^{\alpha}_{\text{ind}}$ by target depth band and site kind (1 is the target's natural level, dashed). Right: the share of circuits that switch the target on to at least half its natural level. Reported for every circuit; not part of the pass criterion.

contexts, where the target is more completely absent, produce a cleaner gap and a more reliable attribution. Semantic closeness and clean causal contrast, in other words, are not the same property, and contrast-context construction should be treated as an experimental variable rather than a solved preprocessing step.

Contrast source for WCM. For weighted circuit masking the source matters less. We refitted the same targets with silence-checked close, random, and distant contrast contexts (Appendix M) and with the store, and scored every fit against every source. Node sets move no more between sources than between two refits with the same source, and the headline faithfulness scores move by a few hundredths; what the source changes is robustness off the training distribution, with each fit most faithful under the contrast-context mean it was trained with. Close contrast contexts are the protocol default (Table 4); [*pending: the train-source × evaluation-source matrix at 288 targets, DAN-64*].

H Specificity

Faithfulness, necessity, and the random-circuit baseline all read the target alone. A circuit can therefore restore its target by lifting the target's whole neighbourhood, its concept, topic, or context

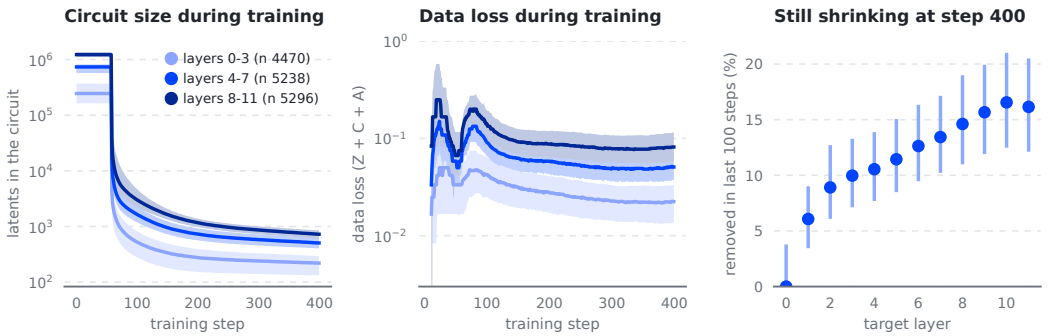


Figure 18: Training dynamics of the 15,004 production fits, by target depth band (median and interquartile range). Left: circuit size (latents above the keep threshold) per step. Middle: the data loss (the Z, C, and A terms, penalty excluded), averaged over each pass through the 48 training contexts (12 steps), since every step sees one batch of 4. Right: the share of members removed over the last 100 steps, by target layer (median and interquartile range).

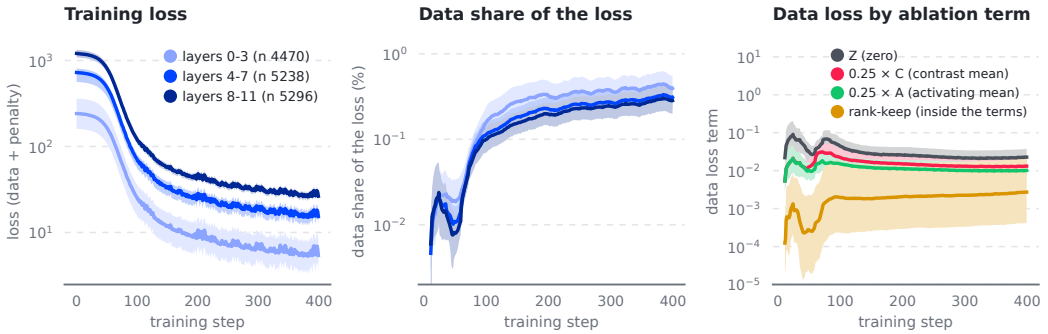


Figure 19: The training loss of the production fits (median and interquartile range; per-pass averages as in Figure 18). Left: the total loss, data terms plus sparsity penalty, by target depth band. Middle: the data terms' share of it; the rest is the penalty on open gates. Right: the data loss split into its ablation terms as weighted in the objective (Z, 0.25× C, 0.25× A), with the rank-keep contribution that each term contains.

type, and pass all three: a concept amplifier rather than a mechanism for that latent. The loss permits it, since reproducing the target's pre-activation constrains the site's stream along one direction of d_{model}, and it leaks into the activation read only when the lifted latents overtake the target in its site's Top-K.

The diagnostic reads the whole code at the target's site, every latent's pre-activation and the SAE's Top-K set at the anchor, in the unmodified model, the empty circuit, and the fitted circuit under each ablation method, on the held-out contexts. It reports the target's rank and Top-K membership; the number of latents lifted above the unmodified model's Top-K cut, against the empty circuit; the overlap of the circuit's Top-K set with the unmodified one; and the faithfulness of *siblings*, latents in the unmodified Top-K at the anchor on at least half of the target's strongest training contexts, computed exactly as the target's, against a control group of the site's other active latents. A circuit is flagged as an amplifier when, under at least one ablation method and on the pre-activation read, its target is restored to at least 0.5 and its median sibling to at least 80% of the target's faithfulness.

In a pilot, most circuits are specific: the target is restored near 1 while siblings and other active latents are restored far less. Under mean ablation, however, circuits lift many latents above the cut that the empty circuit does not, a minority of circuits are amplifiers that pass every faithfulness score,

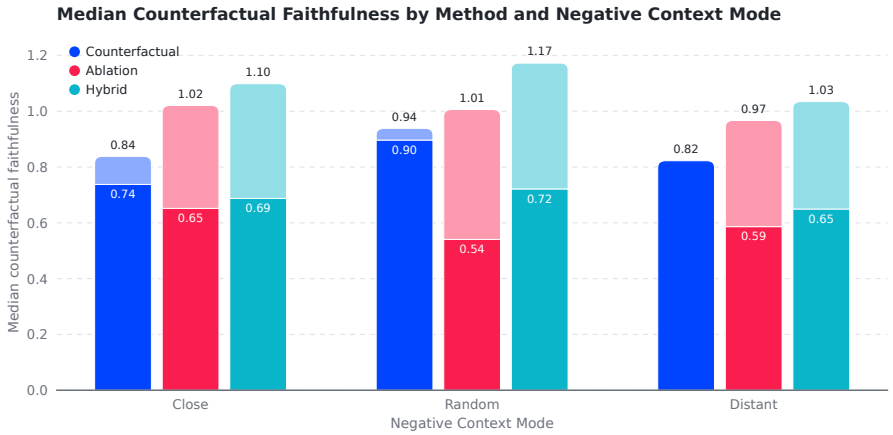


Figure 20: Median sufficiency to induce by discovery method and contrast-context source. Solid segments are medians over each method–source group's accepted circuits; light segments extend to the median of the group's 44 highest-scoring circuits, 44 being the smallest accepted count among the nine groups, for an equal-population comparison. Sources are *close* (the retrieval store of Appendix M, not checked for silence), *random* (uniformly sampled sequences), and *distant* (sequences maximally dissimilar to the activating contexts); this grid predates the silence-checked selection that WCM uses.

and several deep attention targets lose their Top-K place to lifted rivals although their pre-activation is reproduced. The rank-keep term (Section 4.3) keeps the target in its Top-K more often but does not reliably remove amplifiers, so specificity is reported for every circuit rather than claimed for the method. The production run confirms the pilot at scale (15,004 circuits): under Z, A, and C the target is restored to a median 0.94, 0.99, and 0.92 of its pre-activation, its siblings to 0.26, 0.46, and 0.41, and the site's other active latents to 0.14, 0.28, and 0.24. Under the two mean ablations, circuits lift a median of 578 and 517 latents above the unmodified Top-K cut, against 5 and 4 for the empty circuit, and the target stays in its Top-K on 84% and 80% of held-out anchors. 30% of circuits are flagged as amplifiers under at least one ablation method (44% of MLP, 30% of residual-stream, and 11% of attention-output targets), including 38% of passing circuits; 29% of all circuits pass and carry no flag.

The loss terms we tried for rank and specificity before adopting rank-keep are in Appendix P.1.

I Case Studies

How the cases were found. We read passing circuits for targets at layer 6 or deeper (2,238 circuits) with tools that list, for each circuit, its members ranked by contribution at the target's anchor (α times the member's mean activation there times its decoder norm), each member's own top contexts and peak tokens, its engagement on the target's activating against contrast contexts, and its first-order logit effect, and that run hand-written sentences through the clean model. Candidates were kept when a minimal pair of sentences separated the target cleanly. Group names come from the members' own contexts and logit effects, not from automatic interpretation.

Validation. Each case is tested three ways. *Knock-outs*: the circuit is re-scored with one named group removed (its members ablated like any non-member), on the held-out strongest contexts with the production scorer, against the same number of members of similar contribution rank (±15 ranks, excluding every named member and a generic core of latents that fire on most text), over 8 random draws. *Probes*: the target's activation in the clean model on hand-written sentences, as a percentage of its mean activation on its held-out strongest contexts. *Specificity*: the diagnostic of Appendix H. Table 5 summarises six cases; the remaining tables are in the repository.

More on the antiquity circuit. Two further measurements qualify Section 5.3. First, a generic core of seven layer-0 and layer-1 latents that fire on most text is unnecessary under the mean ablations but essential under zero ablation (removing it takes Z to 0.00, A 0.99, C 0.88): with everything else zeroed, these latents keep the stream within the range the later layers expect. Second, run alone on the template sentences under contrast-mean ablation, the circuit keeps ancient Greece and Athens (27) and Rome (18) well above other places (6–8, from Egypt to medieval France; the empty circuit gives 0) but separates dates only weakly (5th century BC 6.9, 19th century 3.8): it was fitted on passages about classical Greece, and its date route is the weaker one, as the knock-outs also show.

What the hunt did not find. No latent we examined tracked whether a stated fact was correct: knowledge appears as latents for entities, eras, and domains, and in the sentences we tried, a wrong date or a false definition activated them as much as the true one, or more.

J Extended Related Work

Masks, scalars, and ablations. Learned masks for circuit discovery descend from subnetwork probing (Cao et al., 2021), applied to computational-graph vertices by Conmy et al. (2023) and to edges by Edge Pruning (Bhaskar et al., 2024) and by uniform gradient sampling with optimal ablation (Li and Janson, 2024), on L_0-gate and continuous-sparsification machinery (Louizos et al., 2018; Savarese et al., 2020). Masks over SAE latents have been learned to localise attributes (Chaudhary and Geiger, 2024), a procedure adopted by benchmarks (Mueller et al., 2025; Karvonen et al., 2025), and to find behavioural circuits by continuous sparsification (Caples et al., 2025); with its coefficients fixed at 1, its objective moved to a single latent, and one mask shared across positions, the latter is our unweighted circuit masking. Every such mask keeps a node at its natural scale or ablates it. The continuous coefficients of Li and Janson (2024) interpolate between the two and play the role of our gates; what they learn besides is a constant for each *removed* component, the dual of our scale for each kept one.

Learned per-unit scales appear with other jobs: scales that double as the pruning criterion (Liu et al., 2017; Huang and Wang, 2018; You et al., 2019); scales for the units kept after pruning, fitted after the mask is fixed (Kwon et al., 2022; He et al., 2017); steering multipliers (Stoehr et al., 2024); sparse fitted weights on SAE latents for a task vector (Kharlapenko et al., 2025); regression weights on feature-to-feature edges (Yin et al., 2026); and manual rescaling of the nodes of a pruned circuit (Gao et al., 2025). Gate-times-magnitude parametrisations separate membership from strength in sparse training and fine-tuning (Louizos et al., 2018; Guo et al., 2021; Lai et al., 2025), where a closed gate means no change or zero rather than an ablated node. Feature-preserving subnetworks have also been pruned by saliency in vision models (Hamblin et al., 2022).

Circuits are routinely evaluated under several ablations (Miller et al., 2024; Mueller et al., 2025) but discovered under one. The nearest objectives sum the clean-to-corrupted and corrupted-to-clean directions of one counterfactual (Chen et al., 2025), or certify a circuit against every patch value in a bounded set (Hadad et al., 2026). Our three-ablation objective can be read as resample ablation over a designed mixture of replacement values, the circuit-level counterpart of attribution averaged over several baselines (Kapishnikov et al., 2019; Erion et al., 2021).

Position-aware circuits. Position handling divides prior work into three strategies, each right for its setting: position-indexed nodes on templatic data and scores summed over positions otherwise (Marks et al., 2025), per-instance graphs built one prompt at a time (Ameisen et al., 2025), and schema-aligned positions across variable-length examples (Haklay et al., 2025). Our node sets keep the position axis with none of these commitments, using per-position selection unioned over each sequence's causal prefix, for the setting none of the three targets: one reusable circuit over arbitrary non-templatic text (Appendix C.3).

Contrast contexts. Retrieved non-activating examples appear in automated interpretability for explanation scoring (Paulo et al., 2024); Paulo and Belrose (2025) find evaluations collapse when

negatives come from neighbouring features, convergent with our random-beats-close result (Appendix G); and O'Neill and Bui (2024) design corrupted negatives for task-level circuit identification. We use retrieved contrast contexts for causal discovery rather than explanation scoring, and treat their construction as a controlled variable with measured consequences.

The SAE basis. Static structure among latents is well documented: co-occurrence and decoder geometry reveal modularity (Li et al., 2025), coactivation-derived components steer behaviour (Deng et al., 2026), meta-decompositions show latents are not atomic (Leask et al., 2025), cross-layer feature-flow graphs track latents through the forward pass (Laptev et al., 2025), though decoder similarity misses most causally strong feature interactions (Droste et al., 2026), and Jacobian SAEs train adjacent dictionaries for sparse computation between them (Farnik et al., 2025). But that structure is correlational, geometric, or weight-based, where the relationships discovered here are context-conditioned and intervention-tested (Appendix F). The basis itself is imperfect: reconstruction errors are pathological (Gurnee, 2024), downstream utility is mixed (Smith et al., 2025), and units are not canonical (Leask et al., 2025). This conditions our claims: reconstruction residuals stay as explicit anchors, evaluation intervenes on measured activations, and the SAE basis is treated as a useful but imperfect sample of the model's internal variables.

K Extended Discussion

The non-identifiability probes. The first five probes below were measured on unweighted circuit masking before scaling coefficients were added, the last two on WCM (Section 5.6). A float32 and a bfloat16 run differ only in rounding, yet exchange 16.5% of nodes at layer 8 (Jaccard 0.835) at unchanged faithfulness and size. Strengthening the sparsity penalty swaps nodes rather than nesting them (containment 0.49–0.80 between consecutive steps); no gate ever freezes, so the stopping step is another sampling axis; 8–25% of nodes sit within 0.05 of the decision cut and are load-bearing; and short-budget and long-budget runs at matched size share only 0.40–0.53 of their nodes. Recomputing the mean-ablation values without a quarter of their contexts, a change that leaves the pass rate and median faithfulness unchanged, moves node sets as far as a change of hardware (median Jaccard 0.66). For a pre-v1 thermodynamics latent, full-budget and quarter-budget circuits both reconstruct the target, yet at 192 and 250 nodes they overlap at only Jaccard 0.60.

Report node sets as equivalence classes. The non-identifiability results (Section 5.6) change what a responsible node-set claim looks like. Set-level and functional claims (size, faithfulness, the cost findings) are stable under every perturbation we applied. Per-latent claims ("latent g belongs to target f's circuit") are not, since node sets for the same target share only 40–85% of their latents: the objective admits an equivalence class of near-equivalent optima, and a single run samples one of them. Claims of the first kind carry this paper; claims of the second kind require survival across perturbations (precision, sparsity path, stopping step) before they are asserted, and the same redundancy that frustrates them is itself a finding: many latents each doing a small share of the same job is exactly what the collective-faithfulness picture predicts.

From circuits to chains. The co-activation results close a loop in the premise: if latent dependencies were recoverable from co-occurrence, latent-targeted circuit discovery could be served by cheap statistics, and they are not. A weighted circuit is one hop: a set of upstream latents that reproduces its target alone, not yet a multi-hop pathway. Because every circuit is indexed by its target, composition is available in principle: splice the circuit that explains an upstream latent into the circuits that use it, then re-validate by intervening at the earliest nodes and measuring propagation through intermediates. Non-identifiability sets its boundary condition: each link is one sample from an equivalence class, so composition must manage classes of node sets, not single ones. That composition-with-validation step, rather than more discovery methods, is the path from dependency neighbourhoods to genuine circuits.

L Pseudocode for the Gradient Methods

Algorithms 1, 2 and 3 summarise the discovery procedures of Appendix C.1 as implemented; Algorithms 4 and 5 summarise restoration-mode attribution (Appendix C.2) and weighted circuit masking (Section 4.3). Thresholds (τ_{act}, τ_{neg}, τ_{sup}, τ_{ind}, τ_{prune}) and top-K sizes are configuration constants shared across the grid; upstream latents additionally require a minimum corpus active count $n_{\min}$, which filters dead or near-dead latents. All forward passes run with identity passthroughs injected at every SAE site (Appendix C.1), so gradients flow to upstream latent activations, and the target-intervention scores ϕ_{ind} and ϕ_{sup} are computed exactly as in Section 4.4. Hybrid fusion (Algorithm 3) unions the two circuits by latent identity, $\mathcal{F}^{\text{hyb}} = \mathcal{F}^{\text{ctr}} \cup \mathcal{F}^{\text{ab}}$, each node keeping its per-source scores and each directed edge inheriting the higher attribution; optional leave-one-out pruning drops nodes with negligible effect on either acceptance metric, and acceptance requires the fused circuit to clear the sufficiency-to-induce and/or necessity thresholds, with the conjunction set per experiment.

Algorithm 1 Contrastive gradient discovery (one target f)

Require: target f; activating contexts $\mathcal{P}_f$; contrast-context source $m \in \{\text{close}, \text{random}, \text{distant}\}$

$\mathcal{N}_f \leftarrow \textsc{ContrastContexts}(f, m)$ ▷ retrieved store, random corpus sequences, or maximally dissimilar sequences

$\mu_f^+ \leftarrow$ mean target activation over $\mathcal{P}_f$

$(\tau_{\text{act}}, \tau_{\text{neg}}) \leftarrow (\tau_{\text{act}}, \tau_{\text{neg}}) \cdot \max(\mu_f^+, 0.1)$ ▷ scale thresholds so focal targets are not penalised

for each contrast context $x \in \mathcal{N}_f$ **do**

  forward pass; capture pre-activation $\hat{a}_f(x, t)$ at the target's site

  $t^* \leftarrow \arg\max_t \hat{a}_f(x, t)$; $\mathcal{L}^{\text{ctr}}(x) \leftarrow \left(\hat{a}_f(x, t^*) - \mu_f^+\right)^2$

  backward pass: $s_g^{\text{ctr}}(x) \leftarrow a_g(x, t_g) \cdot \partial\mathcal{L}^{\text{ctr}}(x)/\partial a_g(x, t_g)$ for every upstream latent g active in x, and $\partial\mathcal{L}^{\text{ctr}}(x)/\partial a_g(x, t_g)$ for one that is absent ▷ raw gradient: $a_g = 0$ there

end for

$S_g^{\text{ctr}} \leftarrow \frac{1}{|\mathcal{N}_f|} \sum_x s_g^{\text{ctr}}(x)$ for every g

$P \leftarrow$ top-K latents by S_g^{ctr} with $S_g^{\text{ctr}} \geq \tau_{\text{act}}$ and active count $\geq n_{\min}$

$N \leftarrow$ top-K latents by $-S_g^{\text{ctr}}$ with $|S_g^{\text{ctr}}| \geq \tau_{\text{neg}}$ and active count $\geq n_{\min}$

$C \leftarrow$ star circuit $\{g \rightarrow f\}$ over $P \cup N$, edge weights S_g^{ctr}

if $\tau_{\text{prune}} > 0$ **then**

  leave-one-out prune nodes whose removal changes ϕ_{ind} by less than τ_{prune}

end if

$(\phi_{\text{ind}}, \phi_{\text{sup}}) \leftarrow$ target-intervention evaluation of C on $\mathcal{P}_f$ and $\mathcal{N}_f$

return C if $\phi_{\text{ind}} \geq \tau_{\text{ind}}$, otherwise reject

Algorithm 2 Ablation gradient discovery (one target f)

Require: target f; activating contexts $\mathcal{P}_f$

for each activating context $(x, t) \in \mathcal{P}_f$ **do**
    forward pass; backward pass from the target's squared pre-activation $\hat{a}_f(x,t)^2$ ▷ `local` mode; path modes use $\hat{a}_f$
    $s_g^{\text{ab}}(x) \leftarrow a_g(x, t_g) \cdot \partial \hat{a}_f(x,t)^2 / \partial a_g(x, t_g)$ for every upstream latent g
end for
$S_g^{\text{ab}} \leftarrow \frac{1}{|\mathcal{P}_f|} \sum_x s_g^{\text{ab}}(x)$ for every g
$P \leftarrow$ top-K latents by S_g^{ab} above threshold, active count $\geq n_{\min}$
$C \leftarrow$ star circuit $\{g \to f\}$ over P, edge weights S_g^{ab}
if $\tau_{\text{prune}} > 0$ **then**
    leave-one-out prune nodes whose removal changes ϕ_{sup} by less than τ_{prune}
end if
$(\phi_{\text{ind}}, \phi_{\text{sup}}) \leftarrow$ target-intervention evaluation of C
return C if $\phi_{\text{sup}} \geq \tau_{\text{sup}}$, otherwise reject

Algorithm 3 Hybrid gradient discovery (one target f)

Require: target f; acceptance mode $\in \{\phi_{\text{ind}}, \phi_{\text{sup}}, \text{both}, \text{either}\}$

$C_{\text{ctr}} \leftarrow$ Algorithm 1 on f; $C_{\text{ab}} \leftarrow$ Algorithm 2 on f ▷ either may reject
if both rejected **then**
    return reject
end if
$C \leftarrow$ fuse returned circuits by latent identity: union of nodes, each retaining its per-source attribution scores; each directed edge keeps the higher attribution
if $\tau_{\text{prune}} > 0$ **then**
    leave-one-out prune nodes with negligible effect on either acceptance metric
end if
$(\phi_{\text{ind}}, \phi_{\text{sup}}) \leftarrow$ target-intervention evaluation of the fused C
return C if the acceptance mode's condition holds ($\phi_{\text{ind}} \geq \tau_{\text{ind}}$ and/or $\phi_{\text{sup}} \geq \tau_{\text{sup}}$), otherwise reject

Algorithm 4 Restoration-mode attribution (one target f; `ig_restoration` substitutes per-round integrated-gradient scoring in line 4)

Require: target f; activating contexts $\mathcal{P}_f$; round budget R = number of upstream sites; per-round admission size k_r

$M \leftarrow \varnothing$; ablate every upstream latent (every latent at its ablation value)
for round $r = 1, \ldots, R$ **do**
    forward with nodes M recomputed from the modified stream and all other latents ablated; capture the target pre-activation $\hat{a}_f$
    score every latent $g \notin M$ by gradient-times-activation at the *current* partially restored stream ▷ re-linearisation: later rounds see the consequences of earlier admissions
    $M \leftarrow M \cup \{\text{top-}k_r \text{ latents not in } M \text{ by score}\}$
    if $\hat{a}_f$ is within tolerance of its natural value **then break** ▷ restoration certificate
    end if
end for
$C \leftarrow$ circuit over M with per-round scores; target-intervention and ablation-faithfulness evaluation as configured
return C

Algorithm 5 Weighted circuit masking, WCM[Z+C+A] with rank-keep (one target f; protocol v1, Table 4)

Require: target f; activating contexts $\mathcal{P}_f$ (train on 32 strongest + 16 mid-band; hold out 16 + 16) and contrast contexts $\mathcal{N}_f$ (48 train / 16 held out); $\lambda = 10^{-3}$; $\gamma_{\mathrm{C}} = \gamma_{\mathrm{A}} = 0.25$; $\gamma_{\mathrm{R}} = 3 \times 10^{-3}$; steps $S = 400$

target $\leftarrow$ natural pre-activation of f at the training anchors (no-grad forward); $\nu \leftarrow$ mean(target^2)

$r \leftarrow \min(K,$ rank of f among its site's other latents) at each training anchor, unmodified model

ablation values: C $\leftarrow$ dense site means over the *training* contrast contexts; A $\leftarrow$ over the *training* activating contexts

initialise $\theta_s = 4$, $\psi_s = \mathrm{softplus}^{-1}(1)$ for every upstream site s

for step $= 0, \ldots, S-1$ **do**

  $T \leftarrow 0.05^{\,\mathrm{step}/(S-1)}$; gates $m_s \leftarrow \sigma(\theta_s/T)$; coefficients $\alpha_s \leftarrow \mathrm{softplus}(\psi_s)$ ▷ geometric anneal $1 \rightarrow 0.05$

  for $\pi \in \{\mathrm{Z}, \mathrm{C}, \mathrm{A}\}$ **do** ▷ same θ, ψ under all three ablation methods

    forward with kept latents at $\alpha\times$ live value, closed latents at their π ablation value

    $\mathcal{L}_\pi \leftarrow$ MSE of f's pre-activation against target; $\mathcal{R}_\pi \leftarrow$ mean relu$(\tau_\pi^{(r)} - \hat{a}_f)^2$

  end for

  AdamW step on $\sum_\pi \gamma_\pi(\mathcal{L}_\pi + \gamma_{\mathrm{R}}\mathcal{R}_\pi)/\nu + \lambda \sum_{s,i} \left[\sigma(\theta_{s,i}) + (1 - \sigma(\theta_{s,i}))\,|\alpha_{s,i} - 1|\right]$ ▷ penalties use $\sigma(\theta)$, not $\sigma(\theta/T)$

end for

circuit $\leftarrow \{\,(i, \alpha_i) : m_i > \frac{1}{2}\,\}$; evaluate on the held-out contexts under Section 4.4

return circuit

Setting	Value
Models and dictionaries	
TuringLLM-1.0-254M	Top-K, $k = 128$, 40,960 latents; attention output, MLP output, residual stream at 12 layers
Pythia-70m	EleutherAI Top-K, $k = 16$, 32,768 latents; the same three sites at 6 layers
GPT-2 small	OpenAI Top-K, $k = 32$, 32,768 latents, layer-normalised input; attention output, residual stream between attention and MLP, MLP output, residual stream at 12 layers
Targets per model	TuringLLM: 455 drawn uniformly at random at every site with an upstream SAE site (layer-0 attention has none), 15,925 in all, restricted to the 15,046 that an earlier run of the pipeline could fit; this drops 879 attention-output targets that were dead or had fewer than 8 activating contexts, so attention sites hold 134–450 targets. 42 more are skipped for fewer than 8 strongest or 4 contrast contexts, leaving 15,004 circuits. [*pending: Pythia-70m and GPT-2 small: 1,024 targets each, one sampling frame, DAN-9*]
Contexts	
Activating contexts	up to 64 strongest and up to 64 mid-band stored contexts, deduplicated (Section 4.1); a store with fewer than 8 is treated as absent. Each pool is split 3:1 into training and held-out contexts: ranked by anchor activation in blocks of 4, block b holds out position $(b + 2) \bmod 4$, so ranks 1–2 always train. With full pools the target trains on 32 strongest (a stratified two-thirds of the 48 strongest training contexts) and 16 mid-band, and holds out 16 strongest (primary) and 16 mid-band (reported). Targets with fewer than 8 mid-band contexts train on all their strongest training contexts (665 of 15,004: 48 for 187, 6–47 for 478); in all, 502 targets train on fewer than 48 contexts and 478 have fewer than 16 held-out strongest contexts
Contrast contexts	64 per target: candidates ranked by highest cosine similarity to any of the target's strongest stored contexts (pooled final-layer representations), its activating contexts excluded; the 8,192 most similar are scanned in order and kept only if the target is outside its Top-K at every position, until 64 are kept (Appendix M); 48 train, 16 held out, stratified by similarity. Targets with fewer than 4 are skipped; 10 have fewer than 64
Anchor	position of the target's peak activation in each context
Weighted circuit masking, WCM[Z+C+A]	
Ablation methods	Z zero ablation (weight 1); C mean over contrast contexts ($\gamma_C = 0.25$); A mean over activating contexts ($\gamma_A = 0.25$), at every layer; both are dense means over the training contexts (no Top-K re-imposed in training)
Objective	MSE to the target's natural pre-activation at the anchors, each term divided by one shared $\nu = \text{mean}(\text{target}^2)$
Rank-keep	$\gamma_R = 3 \times 10^{-3}$ on $\text{relu}(\tau^{(r)} - \hat{a}_f)^2$ inside every ablation term, r the target's rank among its site's other latents at the training anchor in the unmodified model, capped at K
Gates	$m = \sigma(\theta/T)$, $\theta_0 = 4$; T annealed geometrically from 1 to 0.05; nodes are $\{m > \frac{1}{2}\}$
Scaling coefficients	$\alpha = \text{softplus}(\psi)$, $\alpha_0 = 1$; closed gates charged $\lambda(1 - \sigma(\theta))\,\lvert\alpha - 1\rvert$ (no temperature)
Sparsity penalty	$\lambda = 10^{-3}$, flat per latent
Optimiser	AdamW, learning rate 0.05 (constant), weight decay 0.05, 400 steps, batch 4
Execution	nodes recomputed from the edited stream; SAE errors preserved; every upstream site edited
Evaluation (Section 4.4), on the held-out contexts	
Reference contexts	the evaluation's A means, the anchor means set by sufficiency to induce, the specificity siblings, and the random-circuit pool are computed on the strongest training contexts (48 with a full pool) for every target, one reference whether or not the target trains on mid-band contexts; C means use the 48 training contrast contexts, as in training
Read	every pass/fail score reads the target's activation (after Top-K), in every term of the ratio; the same scores on the pre-activation are reported as a diagnostic; training uses the pre-activation, which keeps a gradient where Top-K censors the activation
Faithfulness	ϕ_π^{free} under Z, dense A, sparsity-preserving A and C
Necessity	$\phi_{\text{sup}}(C) = (a_{\text{pos}} - a_{M\setminus C})/(a_{\text{pos}} - a_\varnothing)$, the circuit as one set, its nodes set to zero; $a_\varnothing$ the zero-ablation empty circuit

Target	What it computes	Minimal pair (clean model)	Story groups removed: Z / A / C (random)	Caveat
9.resid.20419 (Section 5.3)	a date or place read as classical Greek antiquity	"In the 5th century BC, the philosopher . . . " 76%; "In the 19th century, . . . " 0%	0.20/0.20/0.19 (0.49/0.98/0.74)	"BC" is partly lexical ("The BC Lions" 44%)
10.resid.15497	the possessive after a name that carries a law, before "law" appears	"Snell's" 94%, "Ohm's" 80%; "Darwin's", "Picasso's" 0%; invented "Zorbel's" 76% in electromagnetism, 0% in painting	0.00/−0.12/0.03 (0.54/0.92/0.64)	amplifier: lifts ∼7,900 latents, target rank 1 → 4
8.resid.31095	"but also" when a sentence attributes causes	"influenced not only by memory but also" 89%; "involves not only memory but also" 0%	0.19/0.41/0.28 (0.73/0.93/0.81)	one latent alone gives most of the drop
8.attn.29991	acid–base dissociation	acid–base sentences 46–87%; other chemistry 0–16%	0.25/0.11/0.16 (0.61/1.19/0.80)	"Her acid wit" 38%
9.resid.7640	the pivot in "not X, but rather Y"	negated clause, then pivot 37–124%; same comma without negation 0–13%	0.00/0.03/0.00 (0.42/0.80/0.39)	fragile: random removals also hurt; lifts 15–21k latents
9.resid.17596	the second "as" of a comparison about importance	"as important for health as" 79%; "as fast as" 0%	importance group no better than random	not validated; lifts ∼30k latents

Table 5: Six case studies from the hunt. Knock-outs remove the case's named story groups together and report faithfulness under zero (Z), activating-mean (A), and contrast-mean (C) ablation; in brackets, the mean over 8 draws of the same number of random members of similar contribution rank. In every validated case, groups named as controls (generic or lexical members) are no worse than random.

M Context Stores and Contrast Contexts

M.1 Context Store Formalism

The collection pass assigns each latent a scored context set $C_f = \{((x,t), a_f(x,t)) : a_f(x,t) > 0\}$. The top-context store retains the K_{top} highest-scoring active contexts,

$$\text{TopCtx}_K(f) = \underset{S \subseteq C_f,\ |S| = K_{\text{top}}}{\arg\text{top}} \sum_{((x,t),s) \in S} s,$$

approximating the upper tail of the conditional activation distribution. The mid-context store is a bounded reservoir sample from the sigma band,

$$\text{MidCtx}_M(f) \sim \text{ReservoirSample}_M\big(\{(x,t) : \mu_f + 0.5\,\sigma_f \le a_f(x,t) \le \mu_f + 1.5\,\sigma_f\}\big),$$

maintained without sorting every active occurrence. Mid contexts distinguish a latent's ordinary operating regime from its most vivid examples and provide probe data where extreme contexts would be too narrow.

Contrast-context retrieval. Let $r(x) \in \mathbb{R}^{1024}$ be the retrieval representation of sequence x: the final layer's residual stream, mean-pooled over positions, captured once during collection, and let $\mathcal{P}_f$ be the sequences stored as activating contexts for f. The query is the normalised centroid of the activating contexts

$$c_f = \text{normalize}\left(\frac{1}{|\mathcal{P}_f|} \sum_{x_p \in \mathcal{P}_f} r(x_p)\right),$$

and the store retains the $n = 64$ sequences most similar to c_f that are not stored activating contexts:

$$\text{ContrastCtx}_n(f) = \underset{S \subseteq \mathcal{D} \setminus \mathcal{P}_f,\ |S| = n}{\arg\text{top}} \sum_{x \in S} \big\langle c_f,\ r(x) \big\rangle.$$

Retrieval is exact rather than approximate: representations are ℓ_2-normalised and full cosine similarities are computed by chunked matrix multiplication on GPU, querying the $K = 512$ nearest candidates before the filter that removes stored activating contexts. This store serves the attribution baselines. Its representation index is a sample of 2×10^5 sequences, so for targets none of whose activating contexts fall in the sample the centroid is undefined, and the store returns one shared default set for all of them (38% of TuringLLM targets); the rest are anchored on the few activating contexts the sample contains.

Contrast contexts for WCM. Weighted circuit masking and its evaluation instead select contrast contexts per target at fit time. Each of the target's strongest stored contexts $\mathcal{P}_f^{\text{top}}$ is represented, read from the index when present and otherwise computed by a forward pass; candidates from the sampled corpus are ranked by their highest cosine similarity to any of them, $\max_{x_p \in \mathcal{P}_f^{\text{top}}} \langle r(x_p), r(x) \rangle$, with all of the target's stored activating contexts (strongest and mid-band) excluded; the 8,192 most similar are then run through the model in rank order, and each is kept only if the target is outside its site's Top-K at every position, until 64 are kept. A target with fewer than 4 kept is skipped (19 TuringLLM targets), and one with 4 to 63 uses what it has (10 targets). The silence check rejects between 14% and 51% of the closest candidates on the targets we inspected, so unchecked retrieval would place many contrast contexts where the target fires. The same selector with random or maximally dissimilar ranking gives the *random* and *distant* sources of Appendix G.

Logit faithfulness. The output-level score of Section 5.5 is the faithfulness formula of Section 4.4 with the target's value replaced by the output metric m, the mean log-probability of the actual next token at the anchor: $(m(C) - m(\varnothing))/(m(M) - m(\varnothing))$, with every in-scope latent outside the circuit mean-ablated.

N Evaluation Protocol Studies

N.1 Why Position Aggregation Fails: Union Coverage

The SAE code is k-sparse per position, so a position-collapsed circuit evaluated across a target's causal prefix must cover the *union* of every position's active set. Measured on a target's probe batch, that union is large at every site (roughly 1–3.7k distinct latents per attention-output site, 7–10k per residual-stream site, and 7–11.8k per MLP-output site), and the burden concentrates in the additive residual carrier: mean-ablating any single residual-stream site alone sends the target to 0.000, while ablating any single attention-output site leaves 0.90–0.96. The harness passes its identity check, keep-all identity at the reduced-precision noise level (0.994), and a selection-agnostic density sweep reproduces the wall; two scoring defects found later in this harness (a mixed pre-/post-Top-K read and ablation limited to member sites) are corrected in protocol v1 (Appendix B), so the numbers here are indicative; position-collapsed faithfulness failure is a property of the dense residual stream plus position aggregation, not an artifact of the harness.

N.2 Sparsity-Preserving Mean Ablation

The classic dense mean ablation assigns every latent outside the circuit its mean, producing a stream with tens of thousands of simultaneously active latents per position, outside the state space a Top-K SAE can produce. The sparsity-preserving variant applies the means and then re-imposes the per-position Top-K (kept latents, then the largest means of the other latents up to k, rest zero), which restores the keep-all identity to exactly 1.0. At scale (32 targets, paired) the sparsity-preserving variant does not systematically shift faithfulness: it raises both a_C and $a_\varnothing$, leaving the ratio a wash (50W/46L). It therefore functions as a cleaner protocol rather than a score inflator; but at small node sets and tight Top-K budgets the two variants can disagree on *rankings*, which is why the body's comparisons name their ablation method.

N.3 Sources of the Mean Ablation

The empty-circuit baseline $a_\varnothing$ is a measured object with its own failure modes. Means over *activating* contexts leak target signal into $a_\varnothing$ increasingly with depth ($a_\varnothing/a_{\text{pos}}$ = 0/23/30/26% at layers 2/8/9/10); means over the target's *contrast* contexts hold $a_\varnothing$ = 0.0000 on every target measured, even under the hardest close contrast contexts. As an ablation method during *discovery*, however, contrast-context mean ablation is neutral for path attribution and destructive for restoration (recomputed faithfulness −0.07 to −0.98, +26% nodes, monotone in contrast-context distance), another instance of discovery and evaluation ablation methods needing to be chosen separately and stated. A three-way cross-tab of mean sources (the target's activating contexts, a random corpus sample, and a farthest-point sample of the corpus by sequence representation) shows method *rankings* are robust to the source while absolute levels shift with how much target signal the mean carries, and sufficiency to induce is invariant to it; every reported number therefore names its ablation method.

O Attribution-Mode and Pruning Studies

Comparisons in this appendix use within-target pairing or depth stratification throughout.

O.1 Recurrence Pruning

A large tail of nodes appears in exactly one probe sequence: context-specific scaffolding rather than mechanism. Pruning nodes that fire in fewer than two probe sequences removes ∼10% of nodes for ∼0.010 recomputed faithfulness at every depth; the harsher eight-sequence bar is smaller but fails at depth (recomputed faithfulness 0.680 vs 1.049 at layer 10, and the dense mean-ablation score goes negative at layer 8), so the two-sequence bar is the smallest setting that survives both ablation methods and depth. Nodes from the contrastive gradient are judged on contrast contexts and nodes from the

ablation gradient on activating contexts, and any node the prune cannot observe is exempted rather than dropped.

O.2 Effect Thresholds Cannot Replace Validated Stopping

Holding the ranking fixed and swapping the stopping rule isolates what does the work. A fixed absolute effect threshold at the field-default value sits ~2.3× above our *maximum* node score (it deletes everything), and even percentile cuts calibrated to our own score distribution crater function (the dose-response of Appendix D), while bisection over the identical ranking preserves it. The stopping rule, not the ranking, carries the validity.

O.3 Selection Rules and the Pruning Frontier

A per-target percentile cut on |attribution| dominates top-N and every per-position–normalised selection rule at equal or better faithfulness with 1.4–2.7× fewer latents (the driving signal is peaked in a *global* magnitude sense; per-position normalisation wastes budget and needs a dead-position guard). The magnitude-bisection prune then composes with it: percentile selection raises the achievable faithfulness ceiling, the prune lowers size at fixed faithfulness, and the composition dominates on all six test targets (layers 3–11). On the mid-depth residual-stream target, top-N selection cannot reach the requested faithfulness at all, while the composed recipe reaches it at less than half the node count.

O.4 Direct-Effect Edges

Edges are attached with the direct-effect construction of Marks et al. (2025): an edge's weight is the indirect effect of an upstream node on the metric via its *direct* effect on the downstream node, with intermediate nodes stop-gradiented, computed as batched vector–Jacobian products. Edges never change the node set or any evaluation score; they are descriptive. But their statistics mirror the discovery mode: local attribution yields sparse shallow graphs (~9 edges per node, depth ~4) where path attribution yields dense deep ones (~40 edges per node, depth ~10), structure following the estimator's linearisation. For weighted circuits the same construction is linearised around the circuit's own run (nodes at their scaled values, every other latent at the ablation value), so the coefficients enter both the upstream change and the downstream Jacobian, and edges are computed separately under each ablation method; a node's out-edges plus its direct edge to the target then sum exactly to its own attribution, which the implementation reproduces to within 1%. The edges also hold beyond first order: with every other node pinned at its circuit value and the upstream node set to its ablation value, the measured change in the downstream node reproduces the edge weight. In a pilot, the wiring is multi-step rather than a star, with most of a circuit's attribution reaching the target through other circuit nodes. It is also dense, with positive and negative routes largely cancelling and no small set of edges carrying most of the weight, and edge sets computed under different ablation methods overlap only partly. We therefore interpret only edges that clear a threshold with one sign under all three ablation methods, and state the share of edge weight they carry; [*pending: wiring statistics on the protocol-v1 circuits, DAN-75*]. An edge's sign combines whether the upstream node raises the downstream one with the downstream node's marginal effect on the target, so a negative edge need not mean suppression. The residual-stream relay of Section 5.5 is such a shared structure.

O.5 Error Nodes for a Target Latent

Keeping SAE errors as ablatable nodes reproduces, for target latents, the error-node picture reported for output targets (Marks et al., 2025), with one asymmetry in our favour: the empty-circuit anchor is error-free here ($a_\varnothing$ identical with and without error terms, 0.000 at layers 2 and 9), so the collapse of the empty-circuit baseline does not arise. Faithfulness at depth, however, is error-carried: zeroing all upstream error terms leaves recomputed faithfulness ≈1.08 at layer 2 but 0.005 at layer 9. Because raw and pruned node sets occupy every upstream site, including error nodes selectively awaits discovery-side error attribution.

P Weighted Circuit Masking Design Ablations

Rejected or superseded designs for WCM, recorded because each closes a natural reviewer question.

P.1 Loss Terms for Target Rank and Specificity

An *off-target* term charging every other site latent for rising above its unmodified and empty-circuit values cut lifts three- to sevenfold at no faithfulness cost, but addressed a property the method need not guarantee (Appendix H) and grew circuits when it also charged harmless lifts below the cut. A *Top-K hinge* requiring only that the target beat the k-th other latent restored faithfulness but, used alone, let circuits keep the target in the Top-K by lifting it together with its neighbourhood. A convex pull towards rank 1 inflated circuits by an order of magnitude at every weight tried, since it never switches off and dominates once the reproduction error is small. The *firing-margin* objective, fitting the target's lead over the Top-K cut, needs its loss normalised by the target's pre-activation scale rather than the margin's, which is near zero for targets at the cut. Rank-keep, which switches off once the target holds its natural rank, is the single term that kept the gains without these costs, and its effect is flat across a thirtyfold range of weights.

P.2 The Sparsity Penalty and Normalisation

Five rejected variants of the sparsity penalty. A gradient-scaled sparsity penalty is *anti*-correlated with need; scaling the penalty inversely to site count is non-monotone; predicting the penalty from target-latent features misses requested sizes by 24–29% where one-probe calibration lands within ∼4%; initialising gates from observed activity buys no speedup (the optimiser updates all parameters in parallel, so deadweight costs nothing); and a gradient-weighted per-site penalty is dominated by a flat one, with the methodological footnote that quantiles over mostly-zero gradient tensors measure sparsity, not scale.

Two further design records support Section 4.3. The normaliser for training under two ablation methods is target-scaled and bounded by construction because its predecessor was not: normalising each term by its own fully-closed-mask loss, which is unbounded far from the data, silently annihilated the zero-ablation term at depth (the two closed-mask losses differ there by up to eight orders of magnitude), reducing WCM[Z+C] training to the C term alone; the design rule is that a normaliser must never be able to annihilate its own term. The one-probe size calibration is the formula $\lambda = \lambda_0 \, (n_{\text{probe}}/n_{\text{req}})^{1/q}$: a single reference run at λ_0 measures n_{probe}, landing requested sizes within a few percent where a per-size search (cf. the sparsity-targeting of Bhaskar et al., 2024) would cost several runs per target.

P.3 Binarisation Sweep

The straight-through estimator makes interior gate values forward-irrelevant, so the sparsity penalty drifts nodes toward just-above-threshold: the decision boundary becomes an *attractor*, and per-step node flips grow through training (218 per step at layer 10), a stopping-step lottery. Annealing instead yields the tightest zero-ablation faithfulness in the programme (0.002–0.10 deviation from unity) at slightly worse mean-ablation scores, another home-turf demonstration. Schedule compression does not help: a 200-step run matches the 400-step metrics at ∼2× the per-step churn (churn is set by per-step pressure, not schedule shape); reach-then-hold leaves churn unchanged and inflates size 21–48%; and truncating a 400-step run at 200 yields node sets 57–72% oversized with the gate still half-soft.

Method	Top-155 ϕ_{ind}	Median ϕ_{ind}	Median nodes	Targets accepted (%)
Contrastive gradient	0.82	0.82	385	40
Ablation gradient	0.97	0.59	193	91
Hybrid gradient	1.07	0.69	264	91

Table 6: Descriptive statistics for the 128-target grid (3 methods × 3 contrast-context sources; 384 runs per method). Median sufficiency to induce (ϕ_{ind}) and node counts are over each method's accepted circuits, pooled across sources. The top-155 column is the median over each method's 155 highest-scoring circuits (155 being the smallest per-method accepted count) for an equal-population comparison; values above 1 are intervention overshoot (Section 4.4). Targets accepted is the share of the method's 384 target–source discovery runs (128 targets × 3 sources) whose circuit passed that method's own intervention gate.

Method	Median ϕ_{ind}	Median ϕ_{sup}	Targets accepted (%)
Sparse co-activation expansion	—	—	0
Differential activation	—	—	0
Ablation gradient (reference)	0.71	1.00	95

Table 7: Correlational discovery baselines under the same 128-target protocol, evidence base, and target-intervention gates as the gradient methods (Appendix F). Dashes indicate that no circuit passed the gates, so no accepted-circuit medians exist. The ablation gradient row is the same-batch reference (121 of 128 targets accepted).

Q Additional Results

Q.1 The Attribution Grid

The *grid*: 128 component-balanced random target latents, each processed by the contrastive, ablation, and hybrid attribution arms (Appendix C) under all three contrast-context sources, for 1,152 discovery runs, summarised in Table 6. Acceptance gates differ by method (Appendix L), so acceptance rates are not comparable across methods. The *panels* behind Appendix D use three to sixteen depth-spanning targets per experiment (layers 2–11, all component kinds); they are directional measurements. Contrastive gradient separates discovery inputs from evaluation inputs by construction; ablation gradient attributes and evaluates on the same probe set.

Q.2 Co-Activation with Fitted Scaling Coefficients: the Strongest Correlational Baseline

Appendix F shows correlational proposals failing the intervention gates outright; the fairest possible strengthening of that challenger is to hand it the weighted-circuit machinery itself. Take the n latents with the largest summed activation at the target's anchors (size-matched to the discovered weighted circuit), and score them twice through the identical pipeline: at natural values ($\alpha = 1$), and with scaling coefficients fitted by the same objective, ablation methods, and budget as the discovered run.

At natural values the selection fails everywhere, and its failure is directional: 0.08–0.51 at layer 2, but 7.6–9.1 at layer 9 — co-activation picks latents that fire alongside the target and misses the rest of the stream that normally holds it in range, so keeping only its picks *overdrives* the target several-fold. With fitted coefficients it becomes the strongest baseline in the programme: zero-ablation faithfulness reaches ∼1 at layer 2 on every target. The joint suite still separates it from the discovered circuits. Under the production recipe, WCM[Z+C+A], best against best, the discovered weighted circuits hold faithfulness under both zero and mean ablation on all five targets (three at layer 2, two at layer 9) (zero ablation 0.92–1.06, mean ablation 0.86–1.18), while the co-activation sets hold both on none: at layer 2 the fitted coefficients calibrate against one background and overshoot the other (mean ablation 1.30–2.47, coefficients reaching 6–10×), and at layer 9 no coefficient assignment reconciles

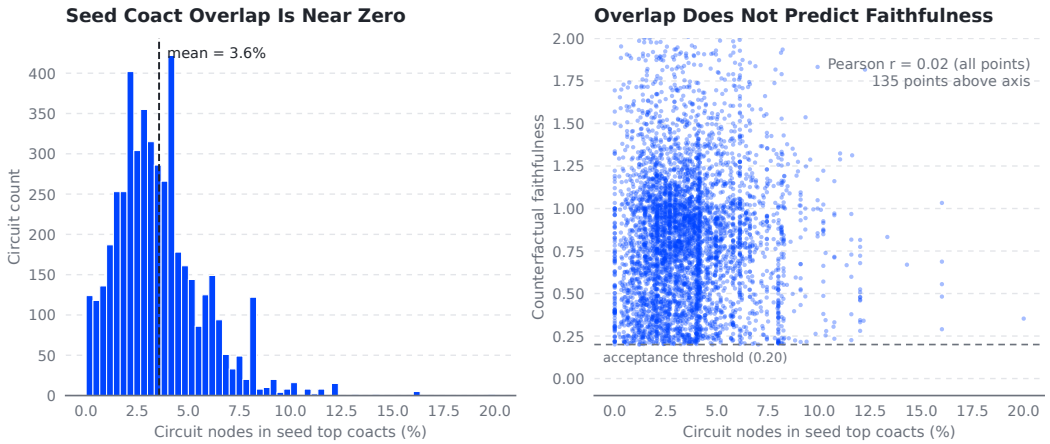


Figure 21: Discovered dependencies are not co-activation neighbourhoods (Appendix F). Left: distribution of the percentage of each circuit's nodes that appear among its target's top co-activation partners (mean 3.6%). Right: the same overlap against the circuit's sufficiency to induce (Pearson $r = 0.02$ over all points; 135 of 4,741 points lie above the truncated axis). The dashed line marks the acceptance threshold (0.20), the minimum sufficiency to induce an accepted circuit can have. Statistics are over the 4,741 circuits with post-analysis co-activation metrics from the 7,525-circuit run.

the co-activation selection with all three ablation methods at once (zero ablation 3.09 on one target, 0.79 on the other). These are TuringLLM targets under the pre-v1 recipe ([*pending: rerun under protocol v1*]). Reconstruction alone is therefore spoofable by a calibrated correlational set, and the revision to Appendix F is precise: correlational *selection* without coefficients fails outright, and with coefficients it fails calibration — what it cannot buy at any coefficient setting is a node set whose contribution is invariant to the ablation method, which is what weighted circuit masking selects for.